电极过程动力学

第三章　电化学步骤的动力学

——单电子反应过程

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3.5 金属电极上电子转移的Marcus理论

唯像动力学（Kinetics）→ 微观动力学（Dynamics）

碰撞理论；过渡态理论；

电子转移理论---Marcus-Hush理论

（1） 表面电化学反应的微观理论发展

（2） Marcus电子转移理论

（3）绝热与非绝热理论

（4）电化学极化与电子能级匹配

（5）隧道效应（电子、核）

（6）电化学界面电荷转移反应的新进展

3.5金属电极上电子转移的Marcus理论

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（1）表面电化学反应的理论发展

1. 唯象理论电极反应动力学理论： k和$\beta \longrightarrow B-V$

不足：不能预示反应物的本质，如溶剂、电极材料以及表面上吸附层的结构对电极反应动力学的影响。

2. 微观理论：能够描述分子结构及其环境如何影响电子转移过程。基于量子力学的界面电荷转移理论。

在均相中的电子转移理论已经形成相当完整的理论体系，但在异相（表面或界面）中的电子转移理论还有待进一步发展。

表面电化学反应的理论发展

Bockris的总结(唯象理论发展过程):

1889年 Arrhenius理论: 化学反应速率与活化能和温度关系

1905年 Tafel关系:电子穿过界面的速率与过电位关系

1924年Butler提出阳极和阴极过程电流和的表示式(β)

1930年Erdey-Gruz和Volmer对Butler公式进行传递系数校正.

1932年Gurney, 1935年Horiuti和Polanyi首次从量子力学研究电化学反应过程.

1933年Frumkin在电极过程动力学公式中考虑双电层性质.

1950年, Conway和Bockris(英国), Gerischer和Vetter(德国)

1955年, Vetter撰写<<电化学动力学>>----→纳米电极体系

J. O'M, Bockris, S. U. M. Khan, Surface Electrochemistry: A Molecular Level Approach, Plenum, 1993. p211-395.

表面电化学反应的理论发展

Bockris的总结(量子理论的发展过程)：

1930年 Oliphant和Moon首次考虑金属/真空界面电子转移反应

1931年Gurney首次从量子力学给出金属中电子穿过能垒到溶液中离子的电子转移速率。这成为此后研究电子转移反应的出发点(Proc. Roc. Soc., 1931, A134, 137) .

1951年Weiss (J. Chem. Phys. 1951, 19, 1066) 电子转移反应中离子能量与外部溶剂区的关系.

1952年Libby, 电荷转移反应中离子溶剂化能的改变.

1956-1975年Marcus, Hush, Levich, Dogonadze, Gerischer的电荷转移理论

1980s Zusman电荷转移理论（溶剂粘度效应）

J. O'M, Bockris, S. U. M. Khan, Surface Electrochemistry: A Molecular Level Approach, Plenum, 1993. p421.

溶液中的电子转移



$$Fe^{3+*} + Fe^{2+} \rightarrow Fe^{2+*} + Fe^{3+}$$

G. Hevesy

1943年获得Nobel奖

因同位素示踪研究化学过程

1945年：

观测到奇怪实验现象：

这是一个很慢的反应, 数小时

未预料到：没有化学键的破坏,仅仅一个电子转移

电子：与核质量相比很轻（$m_H = 1836\ m_e$)

电子质量：9.10938188(72) 10^{-31} kg

质子质量：1.67262158(13) 10^{-27} kg

电子转移交叉反应

$$Fe^{2+} + Ce^{4+} \longrightarrow Fe^{3+} + Ce^{3+}$$

非常不同的反应速率，

反应速率相差15个数量级

$Co(NH_3)_6^{2+} + Co(NH_3)_6^{3+}$ $10^{-6}\ M^{-1}s^{-1}$

$ML_3^{2+} + ML_3^{3+}$ $10^{6}\text{-}10^{7}\ M^{-1}s^{-1}$

L = bpy or 1-10,phenanthroline

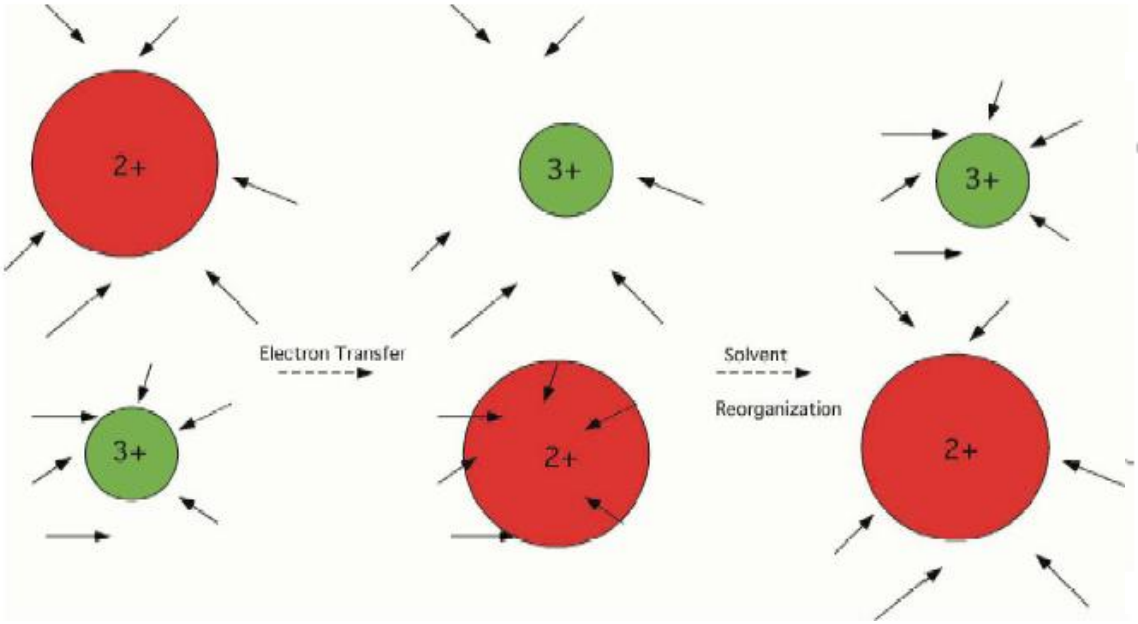
Me = Fe, Ru, Os

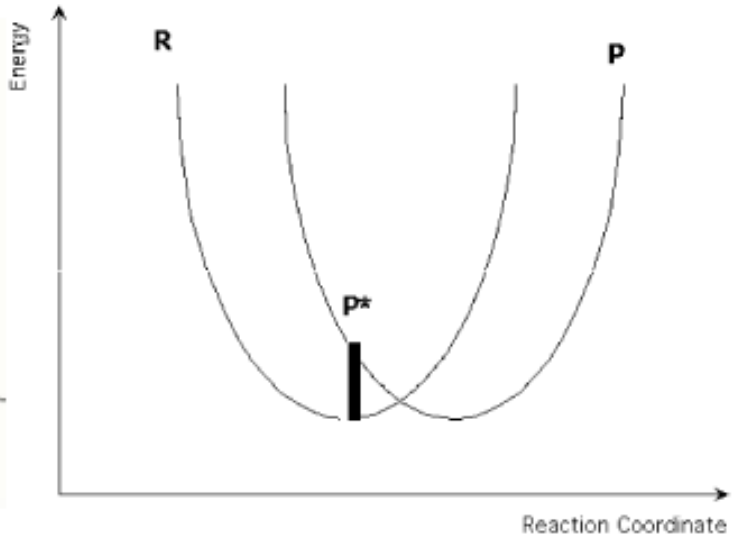
为什么有这么大的差别？

Libby's Basic Idea (1953) $Fe^{3+}* + Fe^{2+} \rightarrow Fe^{2+}* + Fe^{3+}$

电子转移之后溶剂发生重组

由于经电子转移后形成的新物种不是现行溶剂条件下最稳定构型，因此核在电子转移时弛豫到稳定态





Electron transfer among the transition elements; the controlling role of the Franck-Condon principle on rates. W. F. Libby, J. Chem. Phys., 1963, 38 420-5；JPC1952,56,863.



Rudolph A. Marcus

主要的工作完成在1955-1965年

$$Fe^{3+*} + Fe^{2+} \rightarrow Fe^{2+*} + Fe^{3+}$$

在无光情况下这个反应的发生存在以下两个问题：

-----能量不守恒

-----在不正常的高能环境下发生电荷转移

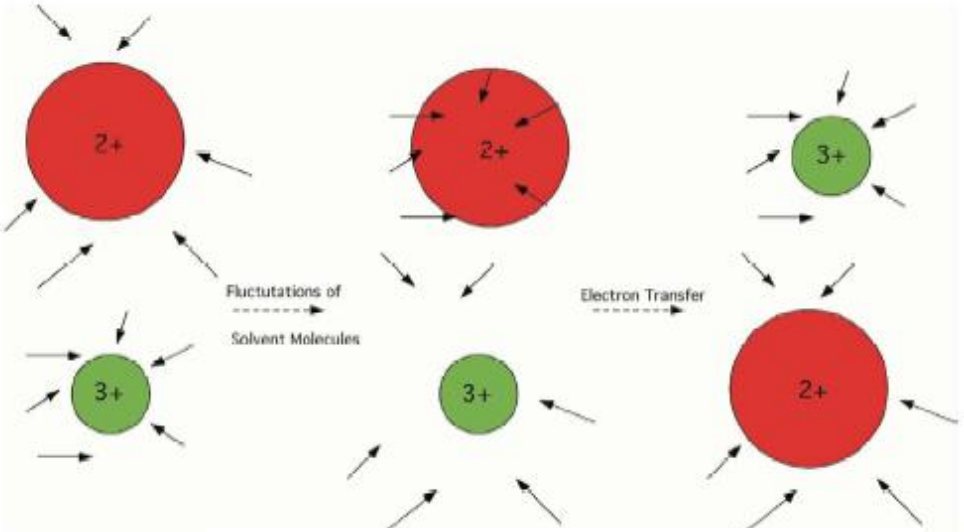
1992年获得Nobel奖

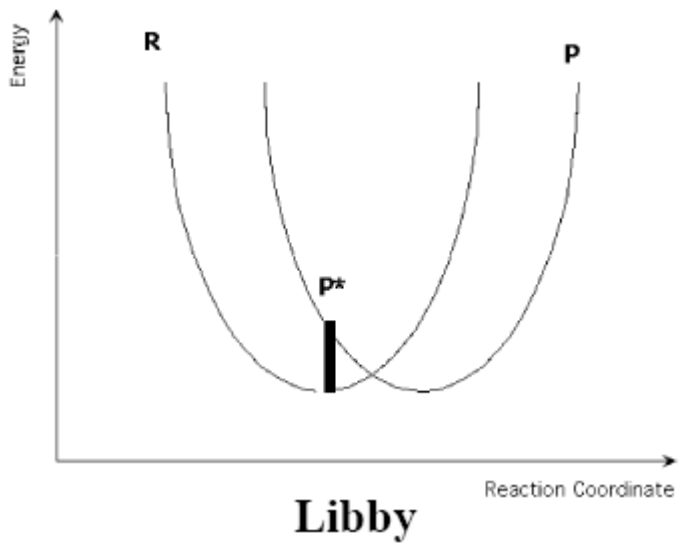
因研究化学体系中电子转移反应理论

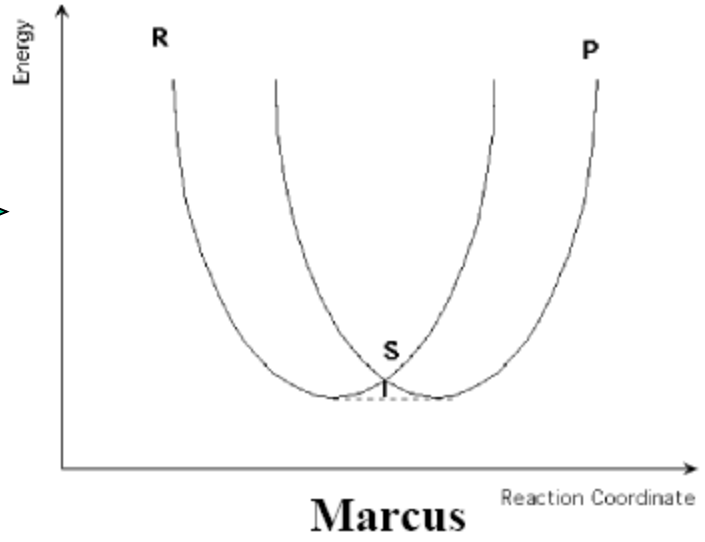
（2）Marcus电子转移理论

Marcus假设： 溶剂结构重组后发生电子转移

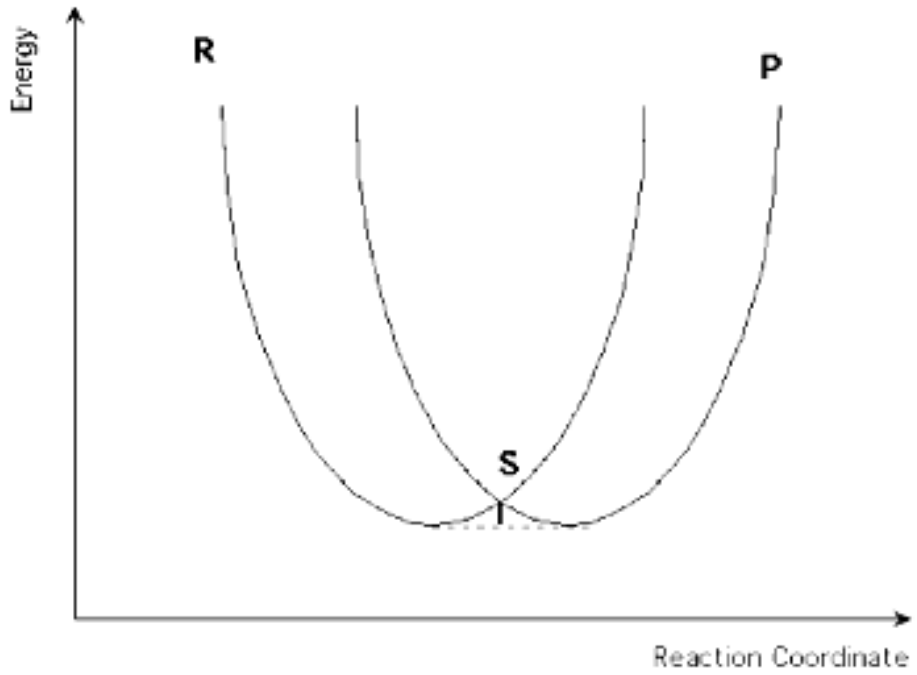
在中心离子最近邻的溶剂分子改变它们的位置，这样增加分子体系的能量。







（2）Marcus电子转移理论



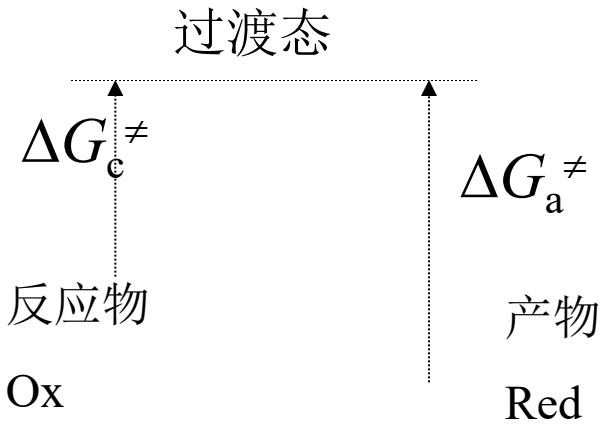
（1）满足Franck-Condon原理，电子转移发生在相同位置和相同原子动量处。

（2）满足能量守恒原理，电子转移前后能量相等。

反应物

反应物

Ox + ne ----- Red

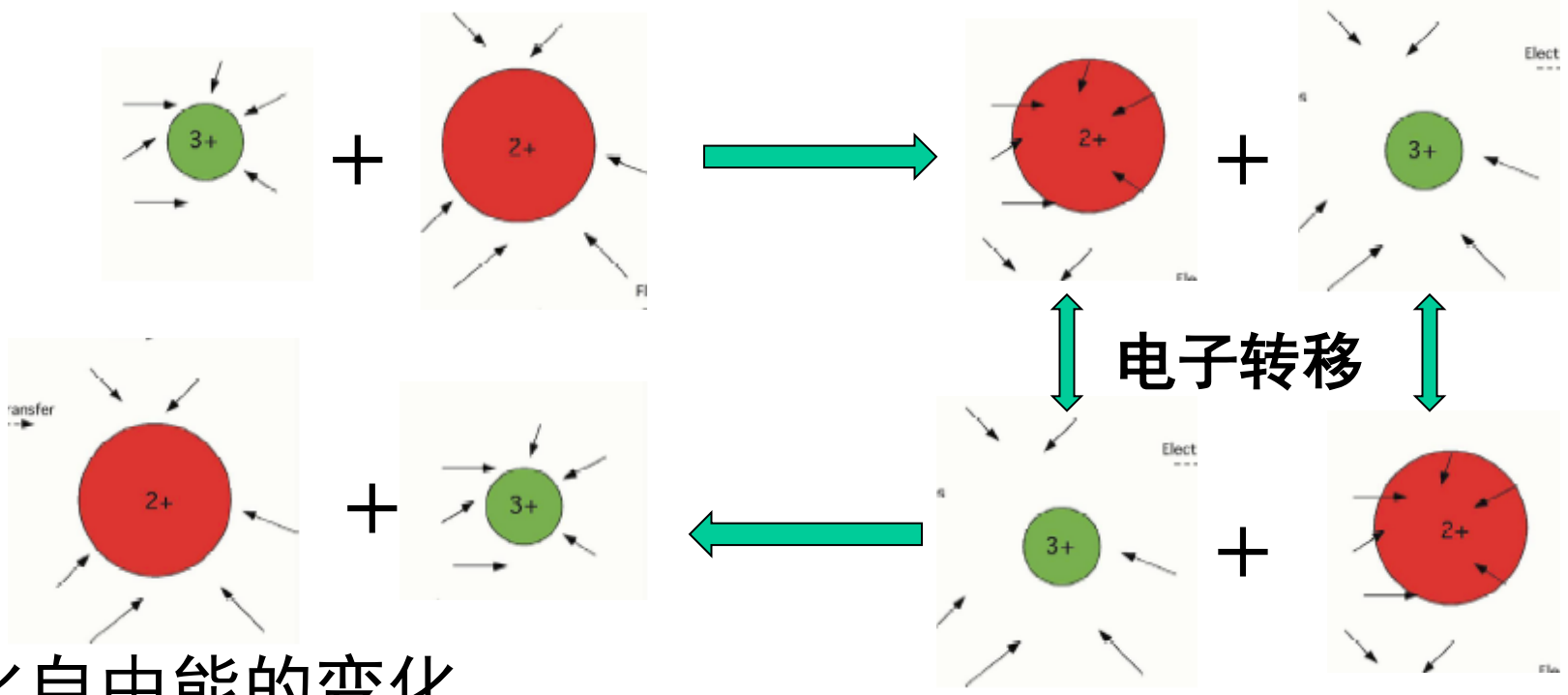


反应坐标

R_{r0} R_{p0} R

如何选择反应坐标
构造反应势能面？

均相中溶液反应 $Fe^{3+*} + Fe^{2+} \rightarrow Fe^{2+*} + Fe^{3+}$

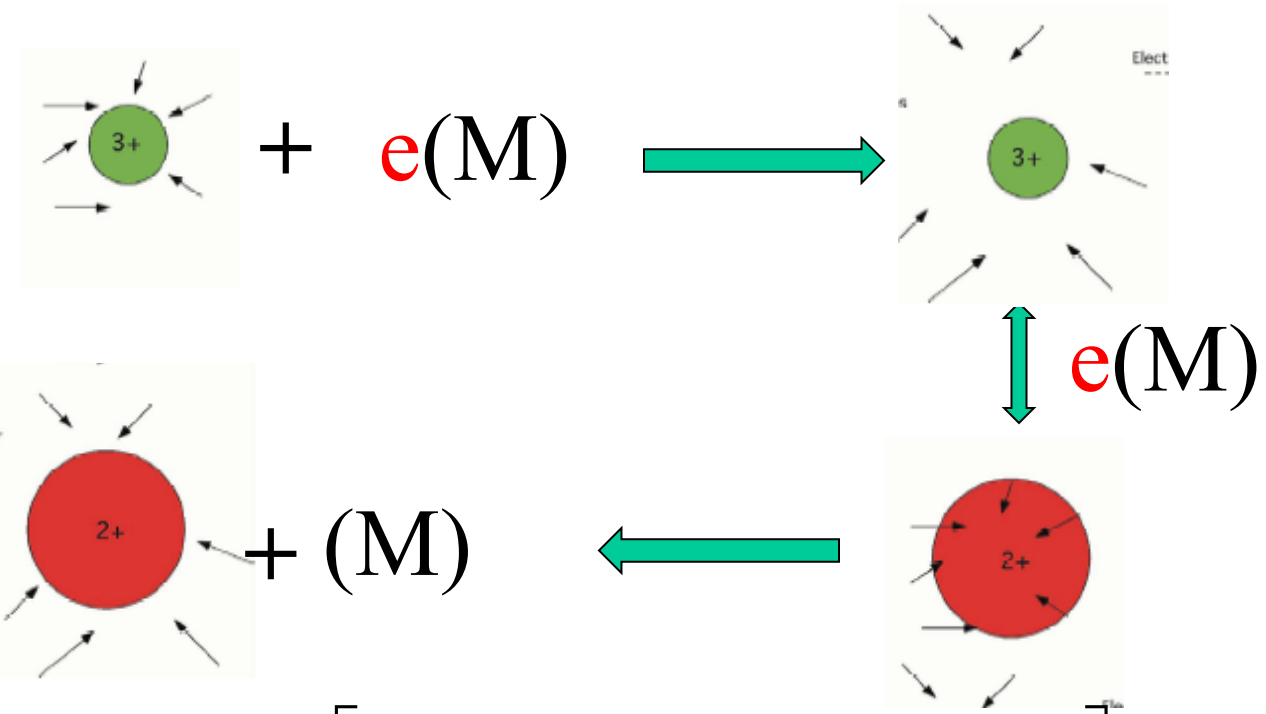


活化自由能的变化

$$\Delta G^{*}_{\text{hom}} = \left[\left(\Delta G^{a}_{Fe^{3+}} + \Delta G^{a}_{Fe^{2+}}\right) - \left(\Delta G^{i}_{Fe^{2+}} + \Delta G^{i}_{Fe^{3+}}\right)\right]$$

A. J. Appleby, J. O'M Bockris, R. K. Sen, B. E. Conway, in Electrochemistry, Vol. 6, Edited by J. O'M Bockris, Butterworths, London, 1973, pp1-40.

电化学反应 $Fe^{3+} + e(M) = Fe^{2+}$



$$\Delta G^{*}_{het} = \left[\frac{1}{2}\left(\Delta G^{a}_{Fe^{3+}} + \Delta G^{a}_{Fe^{2+}}\right) - \Delta G^{i}_{Fe^{3+}}\right]$$

$$\Delta G^{*}_{het} = \left[\frac{1}{2}\left(\Delta G^{a}_{Fe^{3+}} + \Delta G^{a}_{Fe^{2+}}\right) - \Delta G^{i}_{Fe^{2+}}\right]$$

在过渡态时各离子的活化自由能相等，

$$\vec{\Delta}G^{a}_{\mathrm{het}} = \left[\frac{1}{2}\left(\Delta G^{a}_{Fe^{3+}} + \Delta G^{a}_{Fe^{2+}}\right) - \Delta G^{i}_{Fe^{3+}}\right] = \Delta G^{a}_{Fe^{2+}} - \Delta G^{i}_{Fe^{3+}}$$

$$\overleftarrow{\Delta}G^{a}_{het} = \left[\frac{1}{2}\left(\Delta G^{a}_{Fe^{3+}} + \Delta G^{a}_{Fe^{2+}}\right) - \Delta G^{i}_{Fe^{2+}}\right] = \Delta G^{a}_{Fe^{3+}} - \Delta G^{i}_{Fe^{2+}}$$

在平衡电极电位时 $\vec{\Delta}G^{a}_{\mathrm{het}} = \overleftarrow{\Delta}G^{a}_{\mathrm{het}} = \Delta G^{a}_{\mathrm{het}}$

由此可以推出均相与界面的活化自由能关系

$$\begin{aligned}\Delta G^{*}_{\mathrm{hom}} &= \left[\left(\Delta G^{a}_{Fe^{3+}} + \Delta G^{a}_{Fe^{2+}}\right) - \left(\Delta G^{i}_{Fe^{2+}} + \Delta G^{i}_{Fe^{3+}}\right)\right] \\ &= \left(\Delta G^{a}_{Fe^{3+}} - \Delta G^{i}_{Fe^{2+}}\right) + \left(\Delta G^{a}_{Fe^{2+}} - \Delta G^{i}_{Fe^{3+}}\right) \\ &= \vec{\Delta}G^{a}_{het} + \overleftarrow{\Delta}G^{a}_{het} = 2\Delta G^{a}_{het}\end{aligned}$$

$$k_{\mathrm{hom}} = A_{\mathrm{hom}} \exp\left(-\Delta G^{a}_{\mathrm{hom}}/RT\right) = A_{\mathrm{hom}} \exp\left(-2\Delta G^{a}_{\mathrm{het}}/RT\right)$$

$$k_{het} = A_{het} \exp\left(-\Delta G^{a}_{het}/RT\right)$$

$$\Rightarrow \frac{k_{het}}{A_{het}} = \left(\frac{k_{\mathrm{hom}}}{A_{\mathrm{hom}}}\right)^{1/2}$$

（2）Marcus电子转移理论

Franck-Condon原理，非辐射电子转移反应只能发生在位能面的交叉区，因为只有在此处才能同时满足核位置不变和动量守恒这两个条件。将Franck-Condon原理用于非辐射的电子转移反应，可得出以下两个推论：

（A）电子转移反应实际上分三步进行：

$$\xi_{R,0} \longrightarrow \xi^*_R \longrightarrow \xi^*_P \longrightarrow \xi_{P,0}$$

通常第一步较缓慢，第一步也被称为克服“Franck-Condon位垒”的步骤。

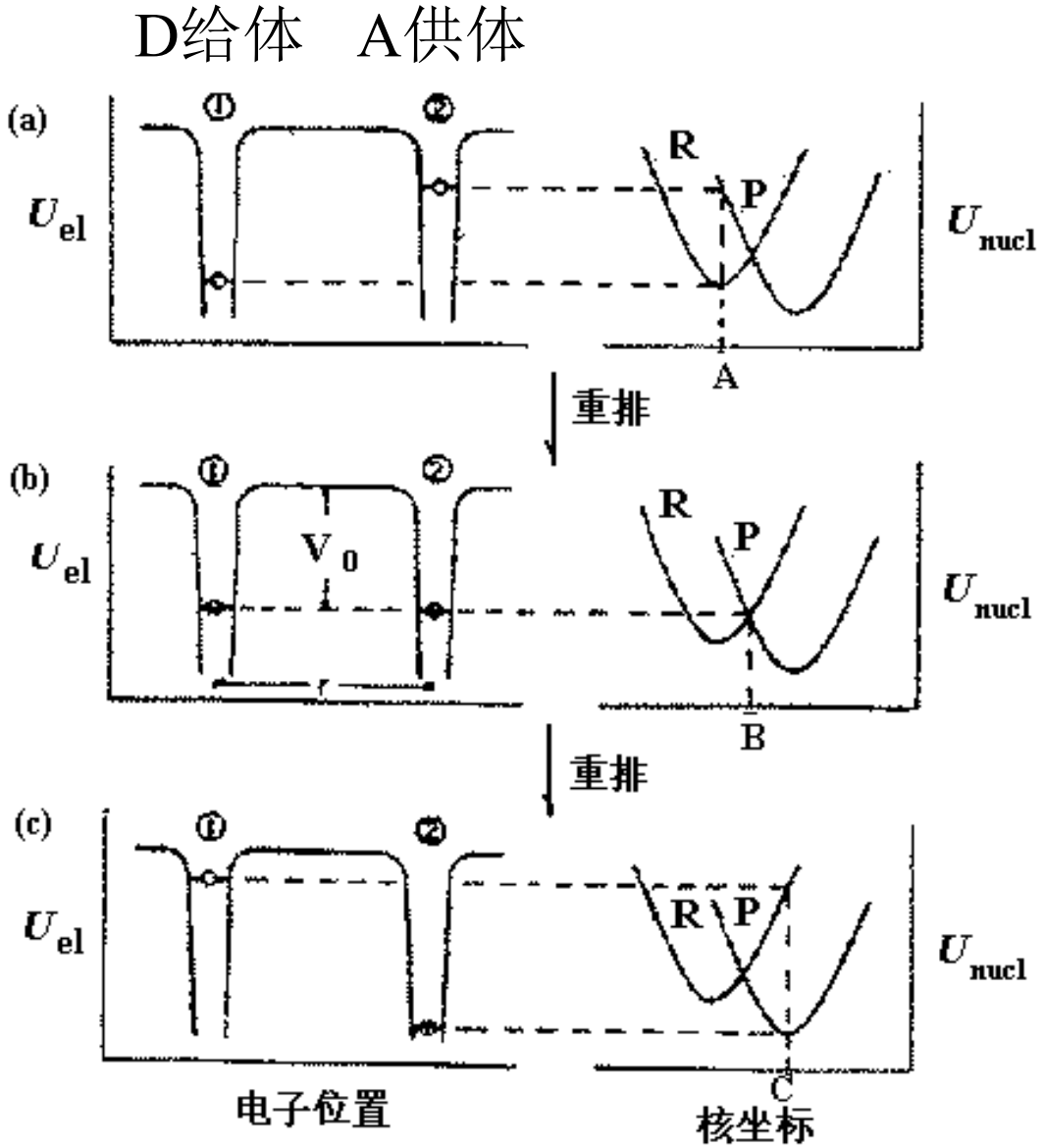
（B）在电子转移反应中，两种活化络合物ξ^*_R和ξ^*_P的构型相同，且在测不准原理的限制范围内具有相等的总能量，不同电子构型。

（2）Marcus电子转移理论

D+A

(DA)*

(DA)



（2）Marcus电子转移理论（如何确定速率常数）

含时Schrödinger方程的微扰理论

$$\hat{H}\psi = i\hbar\frac{\partial\psi}{\partial t}$$

$$\hat{H} = \hat{H}_0 + \hat{H}'$$

它的定态和含时Schrödinger方程

$$\hat{H}_0\varphi_n(q) = E_n\varphi_n(q)$$

$$\hat{H}_0\psi_n^0(q,t) = i\hbar\frac{\partial\psi_n^0(q,t)}{\partial t}$$

$$\psi_n{}^0(q,t) = \varphi_n(q)\exp(-\frac{iE_nt}{\hbar})$$

该波函数是归一化正交完备基

用展开定理, 微扰波函数展开为

$$\psi(q,t) = \sum_n C_n(t)\psi_n{}^0(q,t)$$

由此可得

$$i\hbar\frac{dC_m}{dt} = \sum_n C_n(t)\left\langle\psi_m{}^0\left|\hat{H}'\right|\psi_n{}^0\right\rangle$$

$$\left\langle\psi_m{}^0\left|\hat{H}'\right|\psi_n{}^0\right\rangle = \int dq\psi_m{}^{0*}\hat{H}'\psi_n{}^0$$

考虑微扰量$\hat{H}'$和相应参数λ,

$$C_m(t) = C_m{}^{(0)}(t) + \lambda C_m{}^{(1)}(t) + \lambda^2 C_m{}^{(2)}(t) + \cdots$$

$$C_n(t) = C_n{}^{(0)}(t) + \lambda C_n{}^{(1)}(t) + \lambda^2 C_n{}^{(2)}(t) + \cdots$$

（2）Marcus电子转移理论（如何确定速率率常数）

用上述微扰系数通过推导

零阶（未微扰方程）：

$$i\hbar\frac{dC_m^{(0)}(t)}{dt}=0$$

一阶：

$$i\hbar\frac{dC_m^{(1)}(t)}{dt}=\sum_n C_n^{(0)}(t)\left\langle\psi_m^0\left|\hat{H}'\right|\psi_n^0\right\rangle$$

二阶：

$$i\hbar\frac{dC_m^{(2)}(t)}{dt}=\sum_n C_n^{(1)}(t)\left\langle\psi_m^0\left|\hat{H}'\right|\psi_n^0\right\rangle$$

三阶：

$$i\hbar\frac{dC_m^{(3)}(t)}{dt}=\sum_n C_n^{(2)}(t)\left\langle\psi_m^0\left|\hat{H}'\right|\psi_n^0\right\rangle$$

含时微扰方法用于微扰是时间函数或与时间无关的情况

各种非辐射过程(能量传递和电荷转移)过程，微扰作用不是时间的函数，费米黄金规律(Fermi Golden Rule) :

$$W_{k\to m}=\frac{2\pi}{\hbar^2}\left|H'_{mk}\right|\delta(\omega_{mk})=\frac{2\pi}{\hbar}\left|H_{mk}'\right|^2\delta(E_m-E_k)$$

$$W_k=\sum_m W_{k\to m}=\frac{2\pi}{\hbar}\sum_m\left|H'_{mk}\right|^2\delta(E_m-E_k)$$

$$W=\sum_k P_k W_{k\to m}=\frac{2\pi}{\hbar}\sum_k\sum_m P_k\left|H'_{mk}\right|^2\delta(E_m-E_k)$$

（2）Marcus电子转移理论（如何确定速率常数）

$$\Psi(t)=\sum C_n(t)\Psi_n^{(0)}(q,t)$$

$$i\hbar\frac{dC_n}{dt}=\sum_n C_n(t)\left\langle\Psi_m^{(0)}\left|\hat{H}'\right|\Psi_n^{(0)}\right\rangle$$

$$i\hbar\frac{dC_m^{(1)}}{dt}=\sum_n C_n^{(0)}e^{it\omega_{mn}}H'_{mn}$$ 一级近似

$$i\hbar\frac{dC_m^{(2)}}{dt}=\sum_n C_n^{(1)}e^{it\omega_{mn}}H'_{mn}$$ 二级近似

如果开始处于k态

$$i\hbar\frac{dC_m^{(0)}}{dt}=0\quad\begin{cases}C_k^{(0)}(t)=1\\C_m^{(0)}(t)=0\end{cases}\quad(m\neq k)$$

$$i\hbar\frac{dC_m^{(1)}}{dt}=e^{it\omega_{mn}}C_k^{(0)}H'_{mk}$$

$$i\hbar C_m^{(1)}(t)=H'_{mk}\frac{1}{i\omega_{mk}}e^{it\omega_{mk}}\Big|_0^t=H'_{mk}\frac{1}{i\omega_{mk}}\left(e^{it\omega_{mk}}-1\right)$$

$$C_m^{(1)}(t)=\frac{H'_{mk}}{\hbar\omega_{mk}}\left(1-e^{it\omega_{mk}}\right)$$

$$i\hbar\frac{dC_m^{(2)}}{dt}=\sum_n\frac{H'_{nk}}{\hbar\omega_{nk}}\left(1-e^{it\omega_{nk}}\right)e^{it\omega_{mn}}H'_{mn}$$

$$=\sum_n\frac{H'_{nk}H'_{mn}}{\hbar\omega_{nk}}\left(e^{it\omega_{mn}}-e^{it(\omega_{nk}+\omega_{mn})}\right)$$

$$=\sum_n\frac{H'_{nk}H'_{mn}}{\hbar\omega_{nk}}\left(e^{it\omega_{mn}}-e^{it\omega_{mk}}\right)$$

$$C_m^{(2)}(t)=\sum_n\frac{H'_{nk}H'_{mn}}{\hbar\omega_{nk}}\cdot\frac{1-e^{it\omega_{mn}}}{i\omega_{mn}}\cdot\frac{1-e^{it\omega_{mk}}}{i\omega_{mk}}$$

$$=\left(\sum_n\frac{H'_{nk}H'_{mn}}{\hbar\omega_{kn}}\right)\frac{1-e^{it\omega_{mk}}}{\hbar\omega_{mk}}$$

$$C_m(t)\approx C_m^{(0)}(t)+C_m^{(1)}(t)+C_m^{(2)}(t)$$

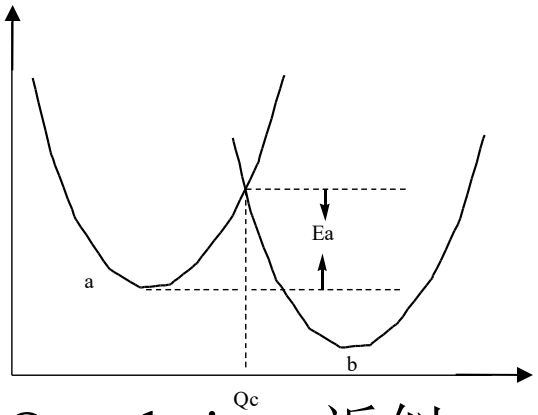
$$=C_m^{(1)}(t)+C_m^{(2)}(t)$$

$$=H'_{mk}\frac{\left(1-e^{it\omega_{mk}}\right)}{\hbar\omega_{mk}}+\left(\sum_n\frac{H'_{nk}H'_{mn}}{\hbar\omega_{kn}}\right)\cdot\frac{1-e^{it\omega_{mk}}}{\hbar\omega_{mk}}$$

$$=\left(H'_{mk}+\sum_n\frac{H'_{nk}H'_{mn}}{\hbar\omega_{kn}}\right)\frac{1-e^{it\omega_{mk}}}{\hbar\omega_{mk}}$$

$$=T_{mk}\frac{1-e^{it\omega_{mk}}}{\hbar\omega_{mk}}$$

（2）Marcus电子转移理论（如何确定速率率常数）



Born-Openheimer近似

$$H'_{b\gamma',a\gamma}=\left\langle \Psi_{b\gamma'}\left|\hat{H}'\right|\Psi_{c\gamma''}\right\rangle=\left\langle \Phi_b\Theta_{b\gamma'}\left|\hat{H}'\right|\Phi_c\Theta_{c\gamma''}\right\rangle=\left\langle \Theta_{b\gamma'}\left|\hat{H}_{bc}'\right|\Theta_{c\gamma''}\right\rangle$$

$$T_{b\gamma',a\gamma}=\left\langle \Theta_{b\gamma'}\left|\hat{H}_{ba}'\right|\Theta_{a\gamma}\right\rangle+\sum_c\sum_{\gamma''}\frac{\left\langle \Theta_{b\gamma'}\left|\hat{H}_{bc}'\right|\Theta_{c\gamma''}\right\rangle\left\langle \Theta_{c\gamma''}\left|\hat{H}_{ca}'\right|\Theta_{a\gamma}\right\rangle}{E_a-E_c}$$

$$=\left\langle \Theta_{b\gamma'}\left|\hat{H}_{ba}'\right|\Theta_{a\gamma}\right\rangle+\left\langle \Theta_{b\gamma'}\left|\sum_c\sum_{\gamma''}\frac{\hat{H}_{bc}'\left|\Theta_{c\gamma''}\right\rangle\left\langle \Theta_{c\gamma''}\right|\hat{H}_{ca}'}{E_a-E_c}\right|\Theta_{a\gamma}\right\rangle$$

$$=\left\langle \Theta_{b\gamma'}\left|\hat{H}_{ba}'+\sum_c\frac{\hat{H}_{ca}'\cdot\hat{H}_{bc}'}{E_a-E_c}\right|\Theta_{a\gamma}\right\rangle$$

$$=\left\langle \Theta_{b\gamma'}\left|\hat{T}_{ba}\right|\Theta_{a\gamma}\right\rangle=\hat{T}_{ba}\left\langle \Theta_{b\gamma'}\left|\Theta_{a\gamma}\right.\right\rangle$$

Condon 近似

$$W=\frac{\left|T_{ba}\right|^2}{\hbar^2}\sqrt{\frac{\hbar^2\pi}{\sum_i S_i\omega_i\hbar kT}}\exp\left(-\frac{\left(\Delta E_{ba}+\sum_i S_i\hbar\omega_i\right)^2}{4kT\sum_i S_i\hbar\omega_i}\right)$$

$$=\frac{\left|T_{ba}\right|^2}{\hbar}\sqrt{\frac{\pi}{\lambda kT}}\exp\left[-\frac{\left(\Delta E_{ba}+\lambda\right)^2}{4\lambda kT}\right]$$

$$\Delta E_a=\frac{\left(\Delta E_{ba}+\lambda\right)^2}{4\lambda}=\frac{\left(\lambda-\Delta G_{ba}\right)}{4\lambda}\lambda=\sum_i S_i\hbar$$

Arrhenius form: $W=Ae^{-\Delta E_a/kT}$

令 $\mathrm{A}=\frac{\left|T_{ba}\right|^2}{\hbar}\sqrt{\frac{\pi}{\lambda kT}}$

活化能 $\Delta E_a=\frac{\left(\Delta E_{ba}+\lambda\right)^2}{4\lambda}=\frac{\left(\lambda-\Delta G_{ba}\right)}{4\lambda}$

（2）Marcus电子转移理论（如何确定速率常数）

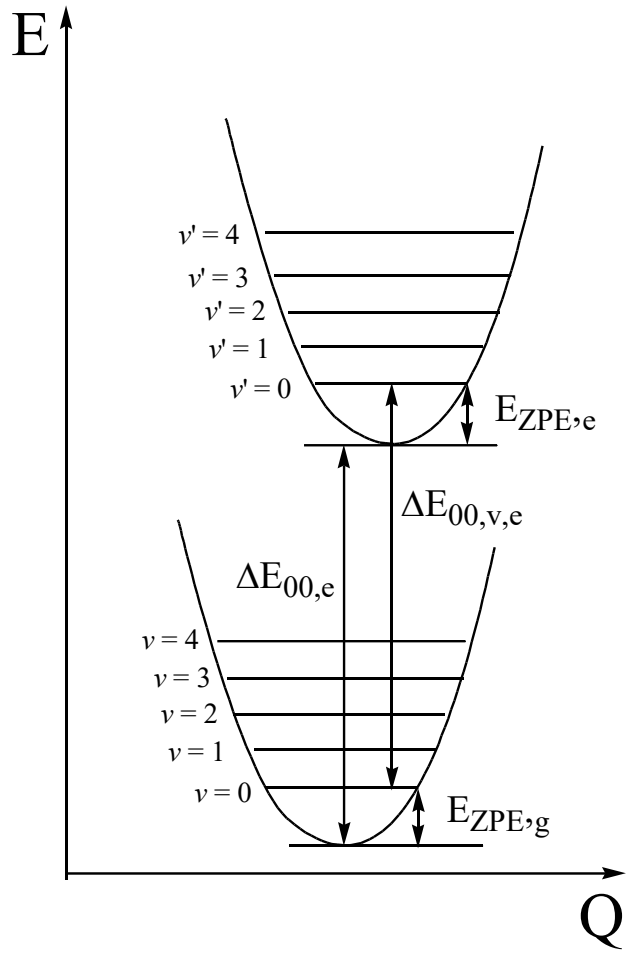
Franck-Condon原理

Franck-Condon原理起先用于解释分子光谱跃迁，后来被用于解释非辐射的电子转移过程。

电子的运动比原子核的运动快得多，在电子转移时核的构型实际上保持不变。

Franck-Condon积分 $\left\langle \Theta_{bv'} \middle| \Theta_{av} \right\rangle$

Franck-Condon因子 $\left| \left\langle \Theta_{bv'} \middle| \Theta_{av} \right\rangle \right|^2$



（2）Marcus电子转移理论（如何确定速率常数）

Fermi黄金律 $k = \frac{2\pi}{\hbar}\left(H_{Ele-mol}\right)^2 (FC)$

$$FC = \sum_h \sum_j \frac{F_j \exp\left[-\frac{\left(\Delta G^0 + jh\nu_h + \lambda\right)^2}{4\lambda_S kT}\right]}{\left(4\pi\lambda_S kT\right)^{1/2}}$$

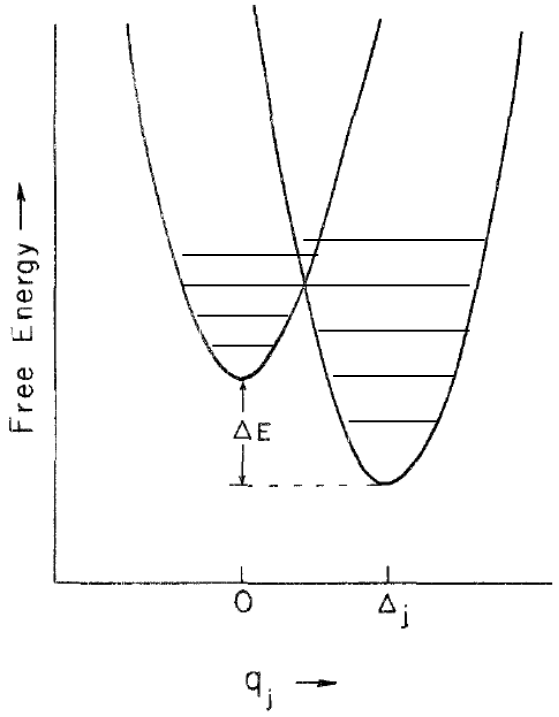
振动模

$h\nu_h > 4k_BT \qquad h\nu_s < 4k_BT$

振动量子数

$$F_j = \frac{S^j \exp[-S]}{j!}$$

$$S = \frac{1}{2}\beta\Delta^2 = \frac{1}{2}\frac{\mu\omega_h}{\hbar}\Delta^2 = \frac{\lambda}{h\nu}$$



J. Phys. Chem., 1974, 78, 2148.

Inorg. Chem., 2001, 40, 1614.

单模(反应坐标)体系且力常数相等

$R = DA,\ P = D^{+}A^{-}$

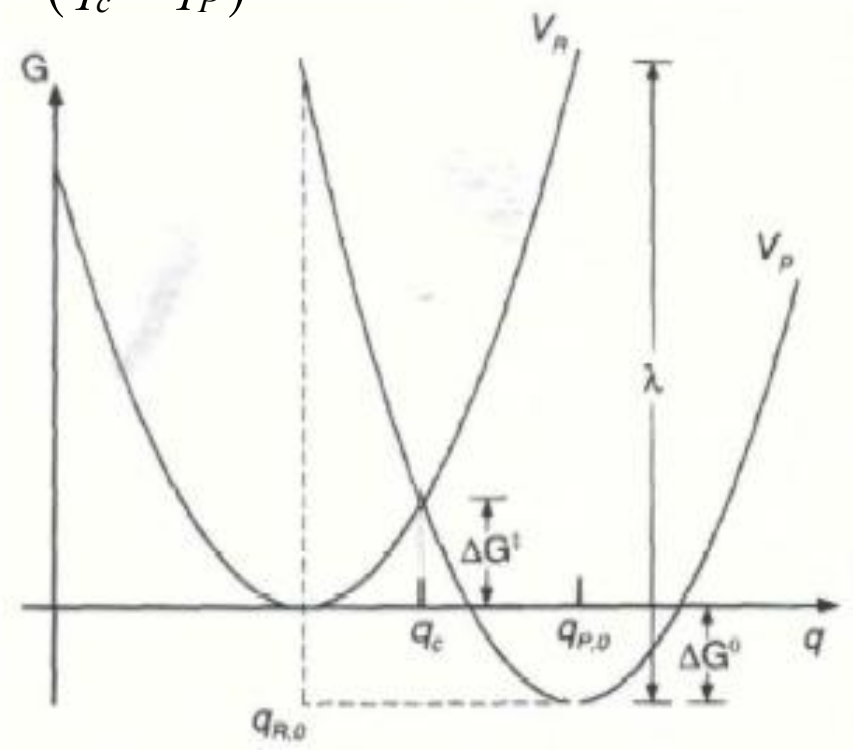
1。确定势能曲线的交叉点(过渡态)

$$V_R = \frac{1}{2}k\left(q - q_R\right)^2 \qquad V_P = \Delta G^0 + \frac{1}{2}k\left(q - q_P\right)^2$$

$$V_R\left(q_c\right) = \frac{1}{2}k\left(q_c - q_R\right)^2 = V_P\left(q_c\right) = \Delta G^0 + \frac{1}{2}k\left(q_c - q_P\right)^2$$

$$kq_c\left(q_P - q_R\right) = \Delta G^0 + \frac{1}{2}k\left(q_P^2 - q_R^2\right)$$

$$q_c = \frac{1}{k}\frac{\Delta G^0}{\left(q_P - q_R\right)} + \frac{1}{2}k\left(q_P + q_R\right)$$



2. 确定内重组能

$$\lambda_f = V_P(q_R) - V_P(q_p)$$

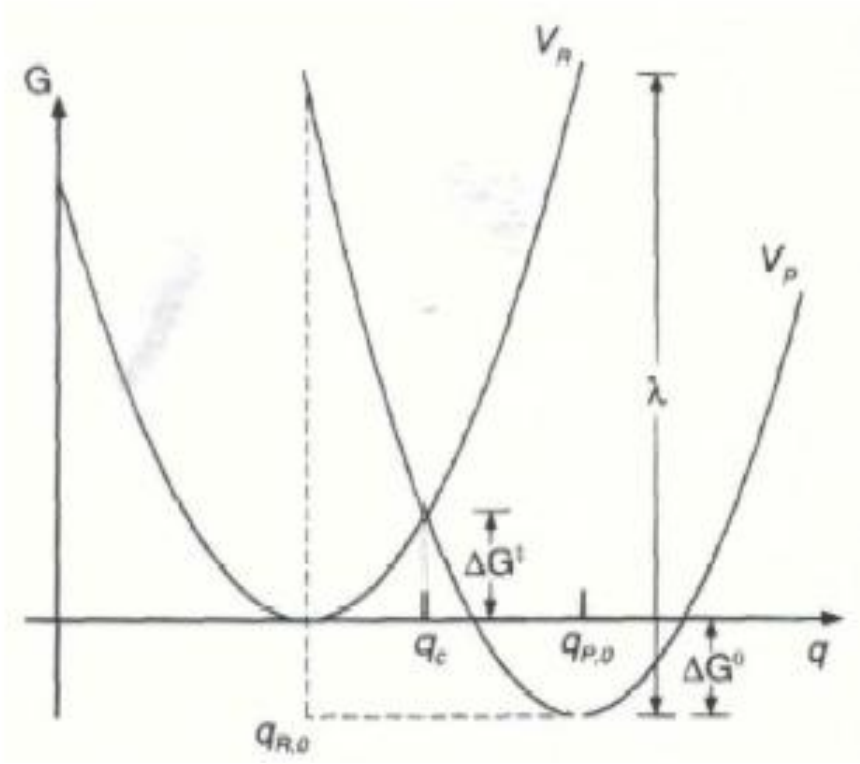
$$= \Delta G^0 + \frac{1}{2}k(q_R - q_P)^2 - \Delta G^0 - \frac{1}{2}k(q_P - q_P)^2$$

$$= \frac{1}{2}k(q_R - q_P)^2$$

λ increases !!!

with the distance $|q_P - q_R|$

with the force constant *k*



Marcus电子转移理论的重组能

重组能λ:

$$\lambda = \lambda_o + \lambda_i$$

λ_o是来自配位球外部的贡献———溶剂重组能；λ_i是内重组能———配位球内部的结构改变，

$$\lambda_i = \sum[\, k_j^R\, k_j^P/(\, k_j^R + k_j^P)](\Delta q_j)^2$$

k_j^R和 k_j^P 是分别是反应物和产物的第j个振动坐标的力常数，Δq_j 是键长的变化。

注意：对溶液中的氧化还原反应，上式中的加和号应同时作用于两个反应物，而对电极反应只涉及一个反应物。

R.A. Marcus, *Faraday Disc. Chem. Soc.*, 1982, 74, 7; R. A. Marcus, N. Sutin, *Biochim. Biophys. Acta*, 1985, 811: 265.

Marcus溶剂重组能

$$\lambda_s = (ne_0)^2\,[(1/(2a_1) - 1/R_\Phi)](1/\varepsilon_{op} - 1/\varepsilon_s)$$

λs 是溶剂重组能，即让离子周围溶剂分子重新排布成为最适宜于电子转移的活化构型所需的能量。

a_i 是包括第一溶剂化层在内的离子半径；

R_{AD}是在电极反应中R_Φ是反应物粒子与其像电荷之间的距离，约为反应物与电极表面之间距离的2倍。

ε_s和ε_{op}是静态介电常数和光学介电常数。

电荷转移在溶液中，r_1和r_2分别是氧化和还原物半径，r_{12}是它们之间的距离

$$\lambda_s = (ne)^2\left[\frac{1}{2r_1} + \frac{1}{2r_2} - \frac{1}{r_{12}}\right]\left[\frac{1}{\varepsilon_{op}} - \frac{1}{\varepsilon_s}\right]$$

$$\lambda_s = (1)^2\left[\left(\frac{1}{2\times 5} + \frac{1}{2\times 5} - \frac{1}{2\times 5}\right)\frac{0.529\,\text{Å}}{1\text{bohr}}\right]\left[\frac{1}{2} - \frac{1}{80}\right] = 0.026\text{au}$$

静电学方法-R. A. Marcus, ONR Tech. Rept. 1957, No. 12, Project NR-051-331; 统计力学方法-R. A. Marcus, J. Chem. Phys., 1965, 43, 679; 标量法静电学方法-R. A. Marcus, in Special Topics in Electrochemistry (Ed. P. A. Rock), Amsterdam: Elesvier, 1979: 180)

3. 确定活化能

$$\Delta G_a = V_R(q_c) - V_R(q_R)$$
$$= \frac{1}{2}k(q_c - q_R)^2 - \frac{1}{2}k(q_R - q_R)^2$$
$$= \frac{1}{2}k(q_c - q_R)^2$$

$$q_c = \frac{1}{k}\frac{\Delta G^0}{(q_P - q_R)} + \frac{1}{2}k(q_P + q_R)$$

$$\lambda_f = V_P(q_R) - V_P(q_p)$$
$$= \Delta G^0 + \frac{1}{2}k(q_R - q_P)^2 - \Delta G^0 - \frac{1}{2}k(q_P - q_P)^2$$
$$= \frac{1}{2}k(q_R - q_P)^2$$

$$\Delta G_a = \frac{1}{2}k(q_c - q_R)^2$$
$$= \frac{1}{2}k\left[\frac{1}{k}\frac{\Delta G^0}{(q_P - q_R)} + \frac{1}{2}(q_P + q_R) - q_R\right]^2$$
$$= \frac{1}{2}k\left[\frac{1}{k}\frac{\Delta G^0}{(q_P - q_R)} + \frac{1}{2}(q_P - q_R)\right]^2$$
$$= \frac{1}{2}\left[\frac{1}{k}\frac{(\Delta G^0)^2}{(q_P - q_R)^2} + \frac{1}{4}(q_P - q_R)^2 + \Delta G^0\right]$$
$$= \frac{1}{2}\left[\frac{(\Delta G^0)^2}{2\lambda} + \frac{1}{2}\lambda + \Delta G^0\right]$$
$$= \frac{1}{4\lambda}\left[(\Delta G^0)^2 + \lambda^2 + 2\Delta G^0\lambda\right]$$
$$= \frac{(\Delta G^0 + \lambda)^2}{4\lambda}$$

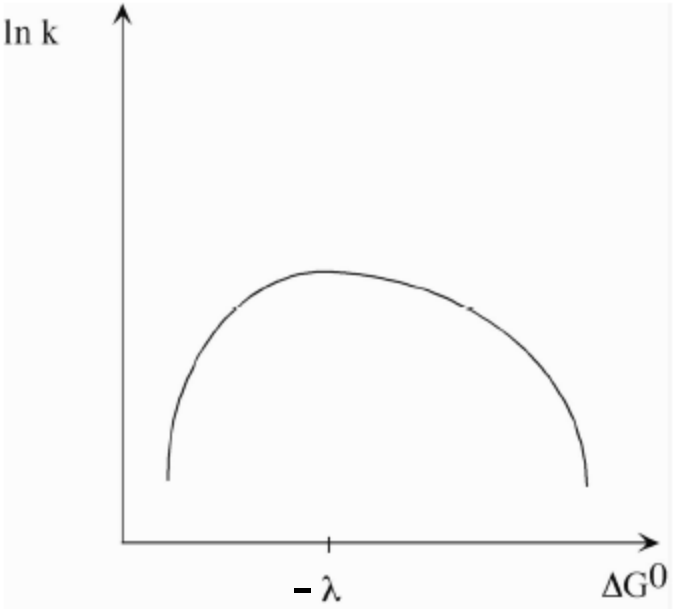
4. 电荷转移速率常数与活化能的关系

$$k = A\exp\left(-\frac{\Delta G_a}{RT}\right)$$

$$\ln k = \ln A - \frac{\Delta G_a}{RT}$$

$$= \ln A - \frac{\left(\Delta G^0 + \lambda\right)^2}{4\lambda RT}$$

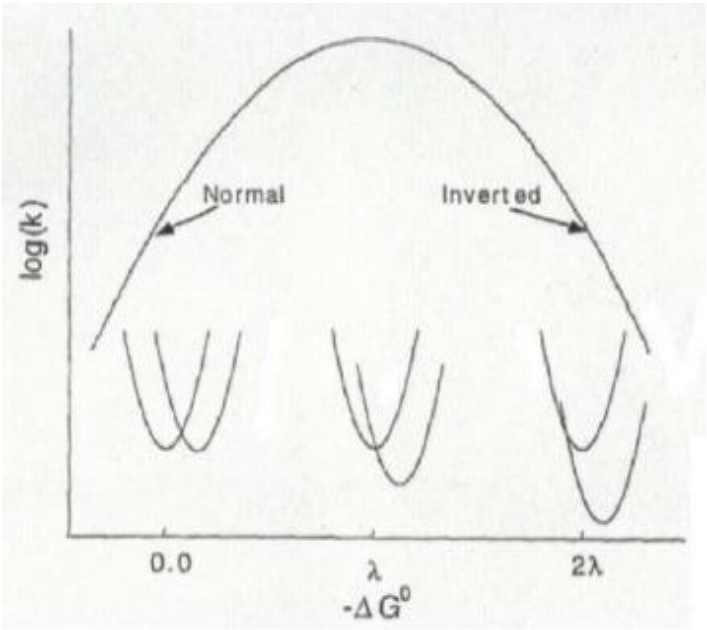
$$= \ln A - \frac{\left(\Delta G^0\right)^2}{4\lambda RT} - \frac{\Delta G^0}{2RT} - \frac{\lambda}{4RT}$$



$\Rightarrow$ 开口向下的抛物线

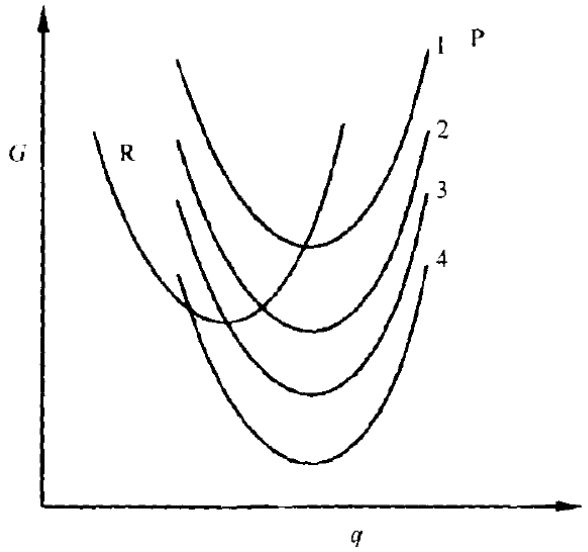
$$\Delta G_a = \frac{\left(\Delta G^0 + \lambda\right)^2}{4\lambda}$$

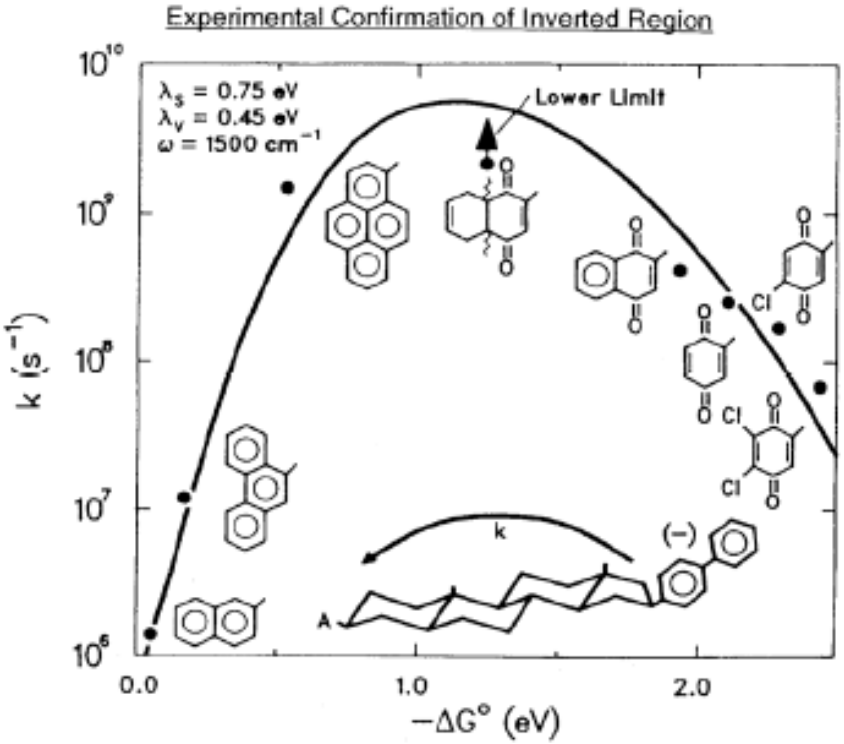
（2）Marcus电子转移理论中k随自由能变化



$$E_a \approx \Delta G^{\#} = \frac{1}{4\lambda}(\lambda + \Delta G^{\circ})^2$$

$$E_a = \frac{(\lambda - |E|)^2}{4\lambda}$$





Miller, *et al*, JACS, 1984,106,3047

（2）电极极化时Marcus电子转移理论

在平衡电极电势下电极反应的

$\Delta G = 0$

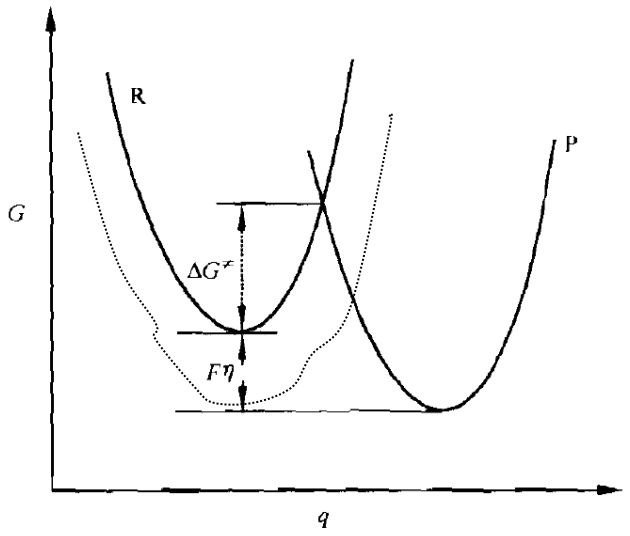
当阴极极化到出现超电势时，

$$k_{red}(\eta) = \frac{2\pi}{\hbar}\int_{-\infty}^{+\infty} f(\varepsilon)\rho(\varepsilon)\sum_k |v_{rk}|^2 \delta(\varepsilon - \varepsilon_k)$$

$$\times\sqrt{\frac{1}{4\pi RT\lambda}}\exp\left[-\frac{(\varepsilon_F - \varepsilon + F\eta + \lambda)^2}{4RT\lambda}\right]d\varepsilon$$

$$= \int_{-\infty}^{+\infty} k_{red}(\varepsilon,\eta)\,d\varepsilon$$

$$\Delta G^* = \frac{(\lambda + \Delta G)^2}{4\lambda} \Longrightarrow \Delta G^* = \frac{(\lambda - F\eta)^2}{4\lambda}$$



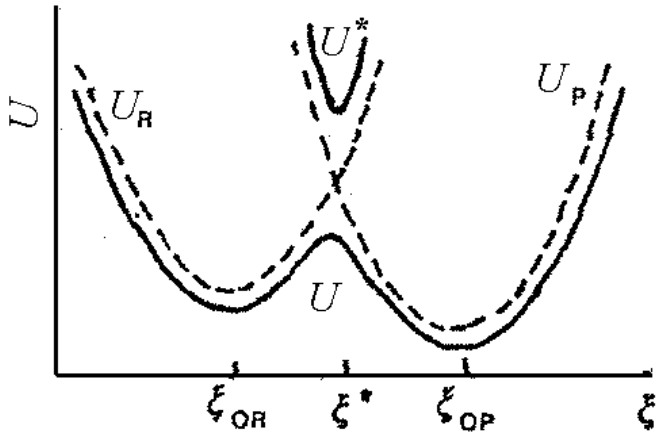
注意：这里引入Marcus理论后，电子子转移速率常数随超电势的变化与B-V方程的变化趋势不同。

A. K. Mishra, D. H. Waldeck, A Unified Model for the Electrochemical Rate Constant That Incorporates Solvent Dynamics, J. Phys. Chem. C 2009, 113, 17904–17914

（3）绝热反应和非绝热反应

绝热反应（adiabatic reaction）和

非绝热反应(nonadiabatic reaction)



U_R和U_P称为反应物和产物的通道位能面（**channel potential energy surfaces**）

U和U^*称为绝热位能面。

绝热位能面在$\xi=\xi^*$处能级分裂：

$$E_U^* - E_U = 2H_{12}$$

$2H_{12}$：相互作用能或共振能。

$$E_U - E_U^* = 2H_{12}$$

$2H_{12}$是体系的相互作用能或称共振能。

如果反应体系从始态向终态的转变只在位能面U上进行，则发生绝热反应，否则发生非绝热反应。

（3）绝热反应和非绝热反应

绝热反应的条件：Landau-Zener理论　绝热跃迁几率P_{ad}

$$P_{ad} \approx 1 - \exp(-\gamma_e) \qquad (2.2.6)$$

$$\gamma_e = 2\pi H^2_{12} / (h|\upsilon| \cdot |F_1 - F_2|) \qquad (2.2.7)$$

γ_e：绝热参数；υ：体系接近跃迁区的速度；F_1和F_2是势能曲线的斜率，表示体系复原为反应物态或维持产物态的能力。

（1）如果H_{12}非常大、以致$\gamma_e >> 1$，则$P_{ad} = 1$，反应是绝热的。因为H_{12}越大，意味着体系由U窜入U^*的机会越小。

（2）若H_{12}不大，但$F_1 \approx F_2$，P_{ad}也可能接近于1。如：$tg\theta_1 = tg\theta_2$，$\beta = tg\theta_1 / (tg\theta_1 + tg\theta_2) = 1/2$。此时，$\beta$值可作为判断反应绝热性的参考依据，$\beta = 0.5$的反应理应是绝热反应。

（3）绝热反应和非绝热反应

经典Marcus电子转移理论

$D + A \longrightarrow (DA)^*$　　$(DA)^* \longrightarrow (DA) \longrightarrow$

Marcus认为决速步骤的速率常数k可表示为

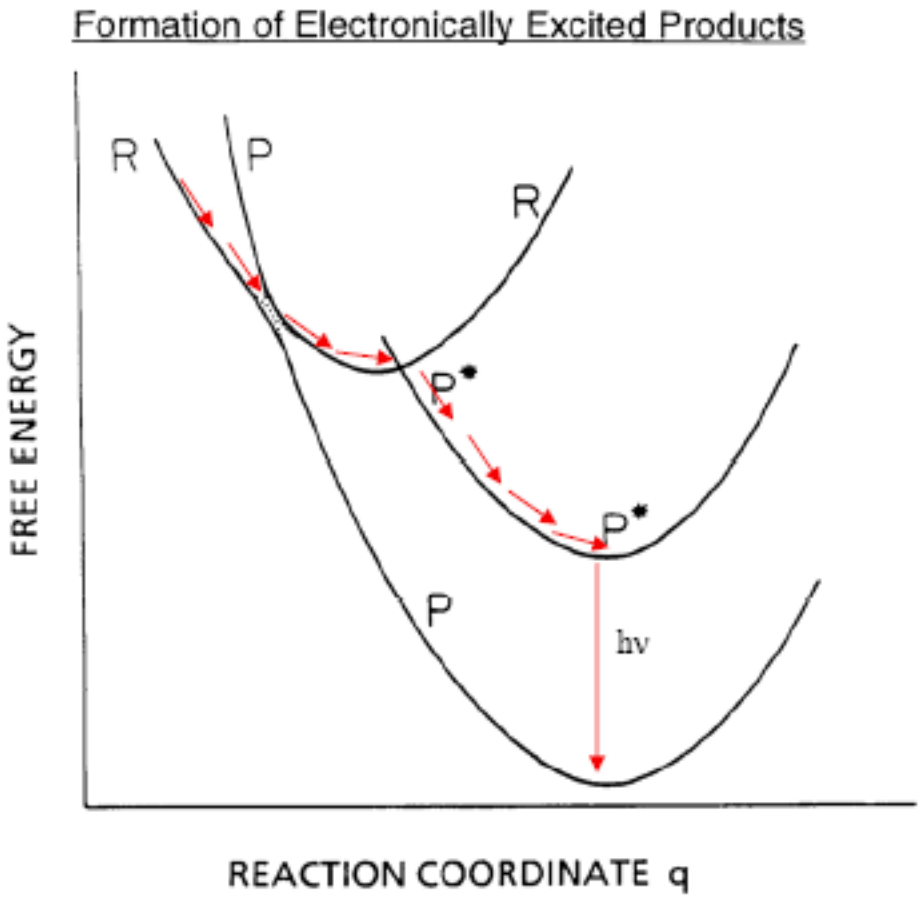
$$k \approx k_1 = \kappa Z \exp\left(-\Delta G^*/k_B T\right)$$

其中κ 是穿透系数(transmission coefficient)

绝热反应 $\kappa = 1$

Hale J M. in *Reactions of Molecules at Electrodes* (ed Hush N S) , New York: Wiley Interscience, 1971: pp. 229

Inverted Region ⇒ Chemiluminescence
Marcus' Prediction of 1965



（4）电化学极化与电子能级匹配

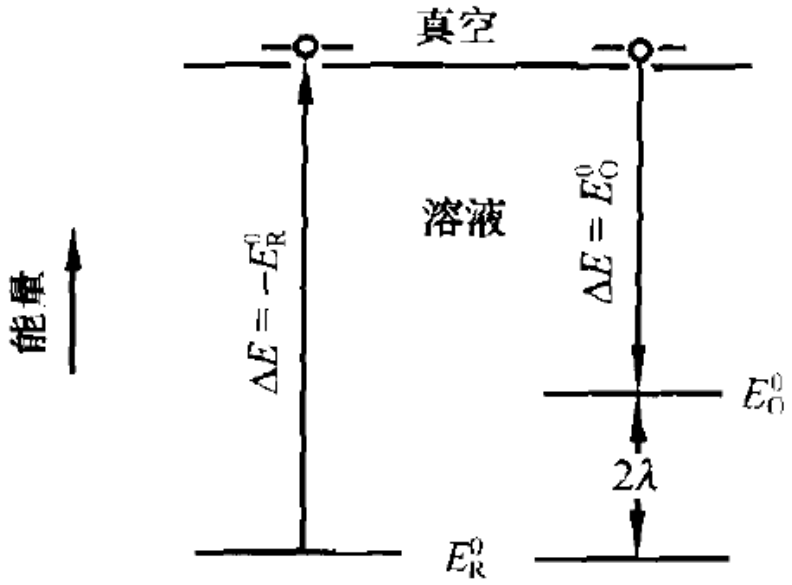
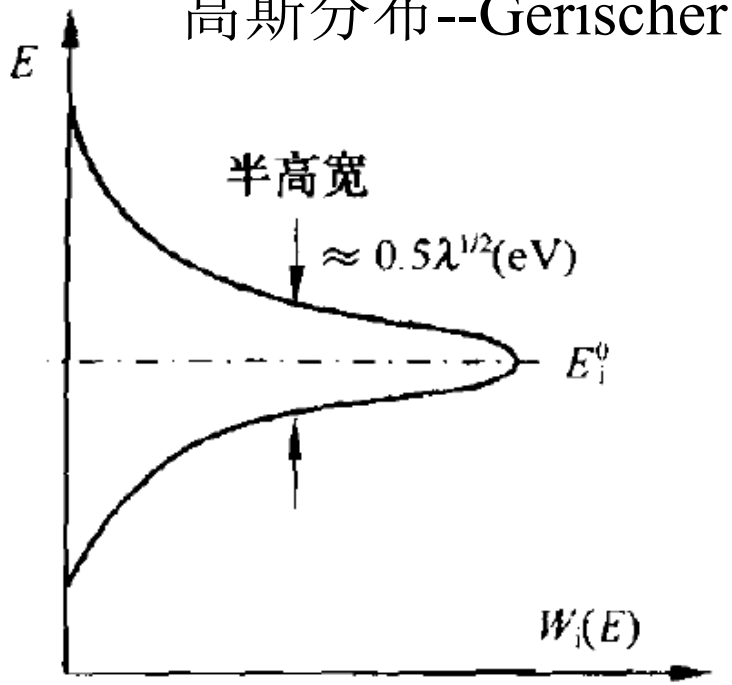


图 4.29　E_R^0 和 E_O^0 的位置

高斯分布--Gerischer



电子态密度分布

$$DOS(E) = (4\pi\lambda kT)^{-1/2} \exp\left[-\frac{(E-E^0)^2}{4\lambda kT}\right]$$

（4）电化学极化与电子能级匹配

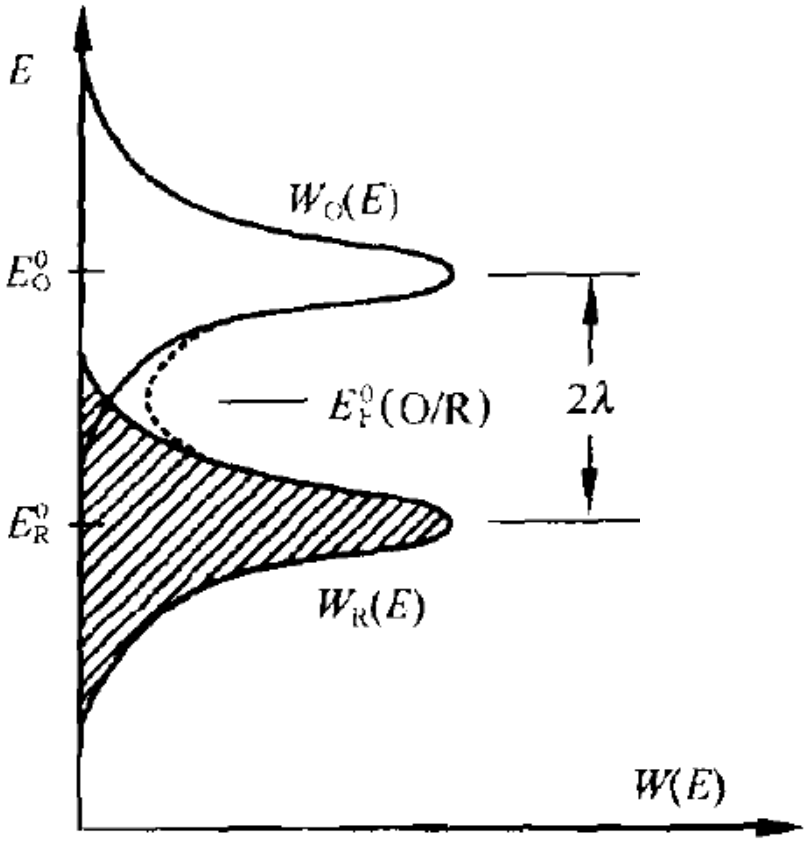
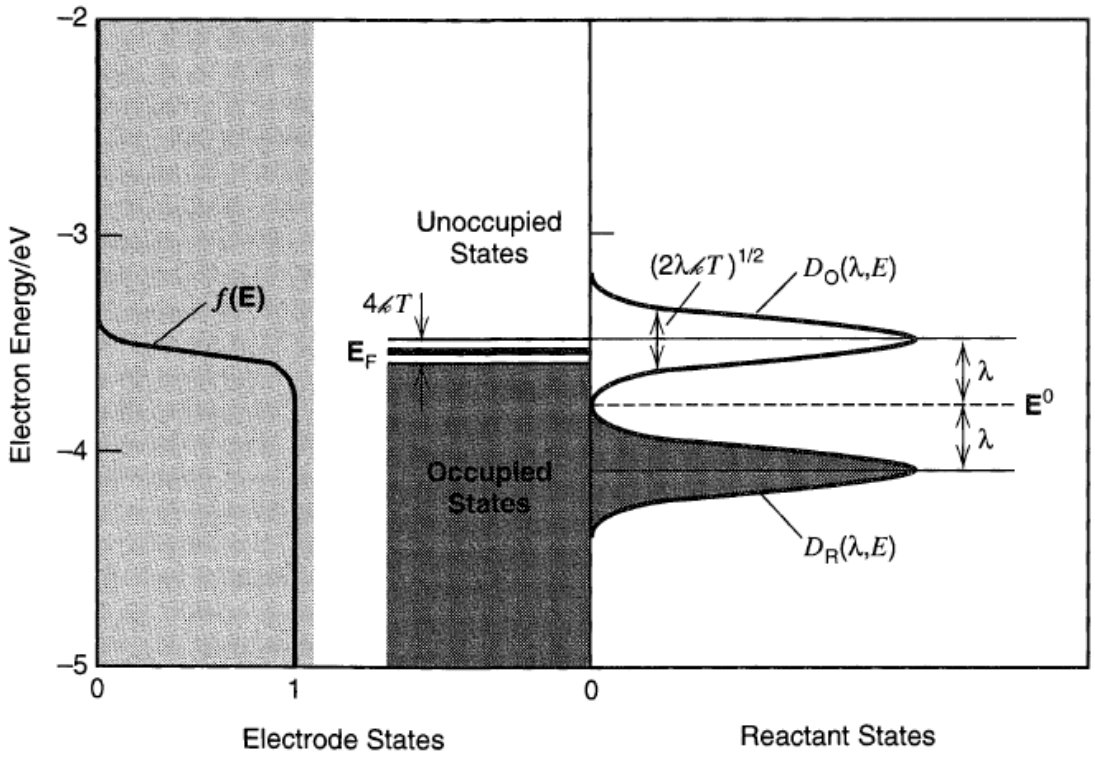


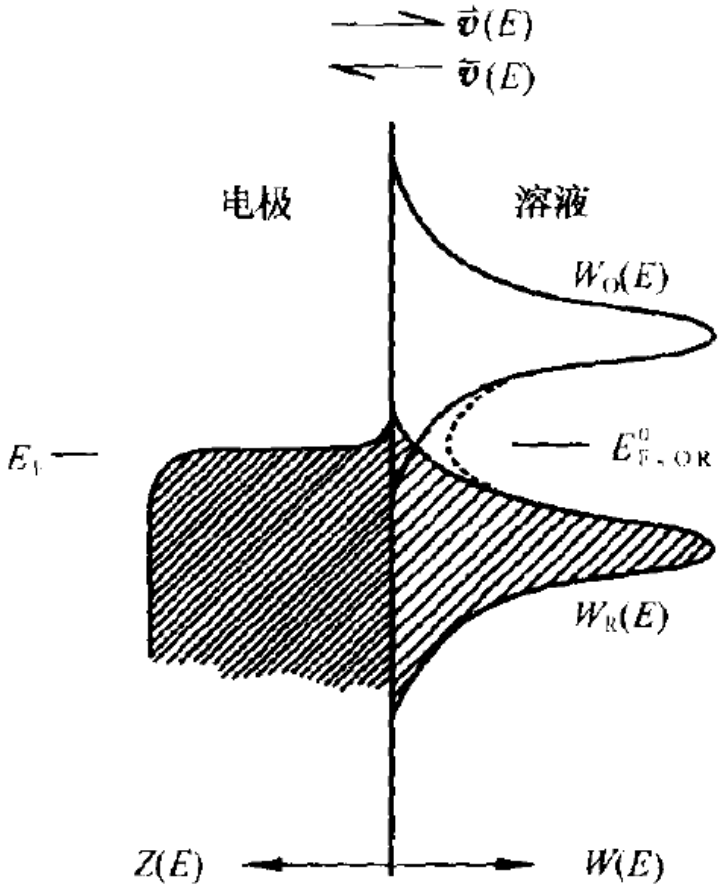
图 4.31　溶液中 O/R 体系价电子的态分布及被电子占据的情况

（4）电化学极化与电子能级匹配

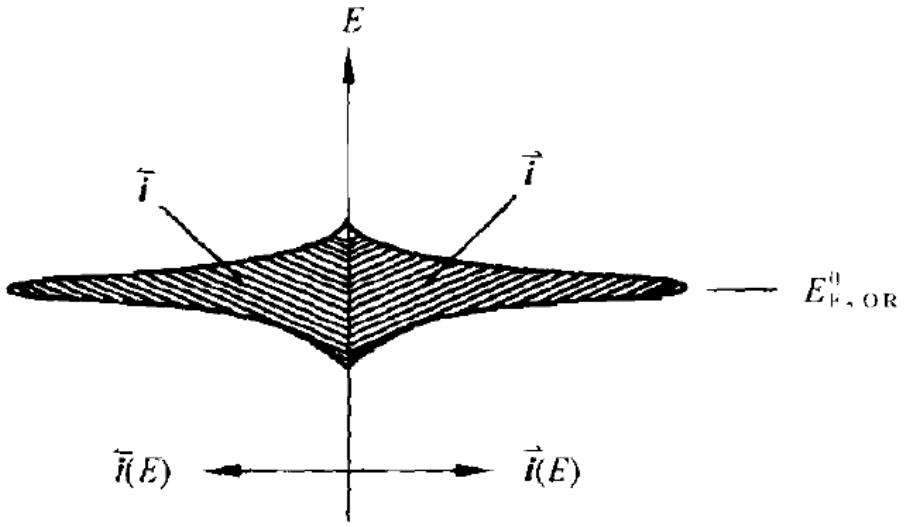


Gerischer电子态模型，H. Gerischer, Adv. Electrochem. Electrochem. Eng., 1961, 1, 139; Physical Chemistry; An Advanced Treatise, Vol. 9A, H. Eyring, D. Henderson, W. Jost, Eds., Academic, New York, 1970.

（4）电化学极化与电子能级匹配

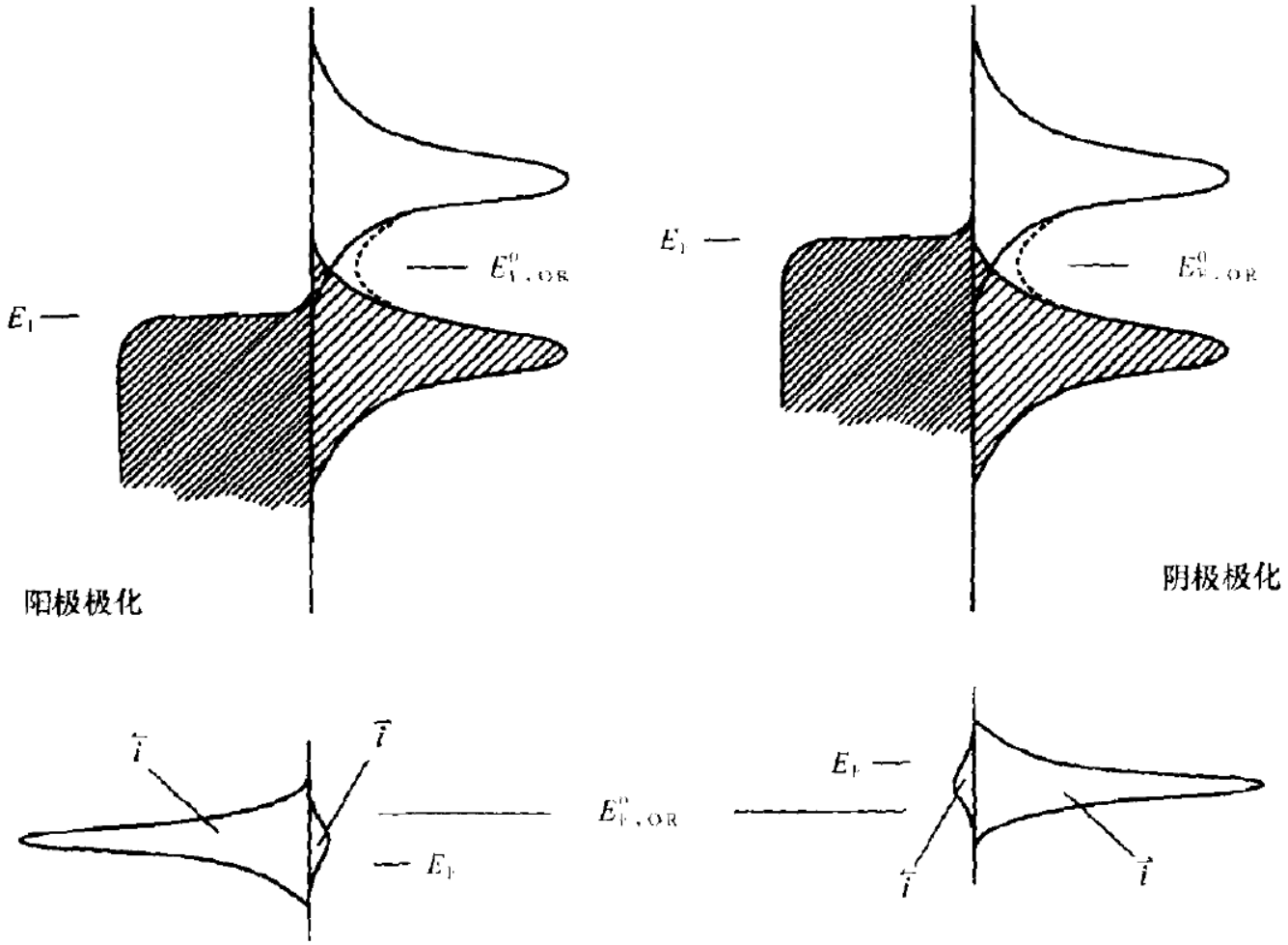


电极/溶液界面两侧的电子态密度



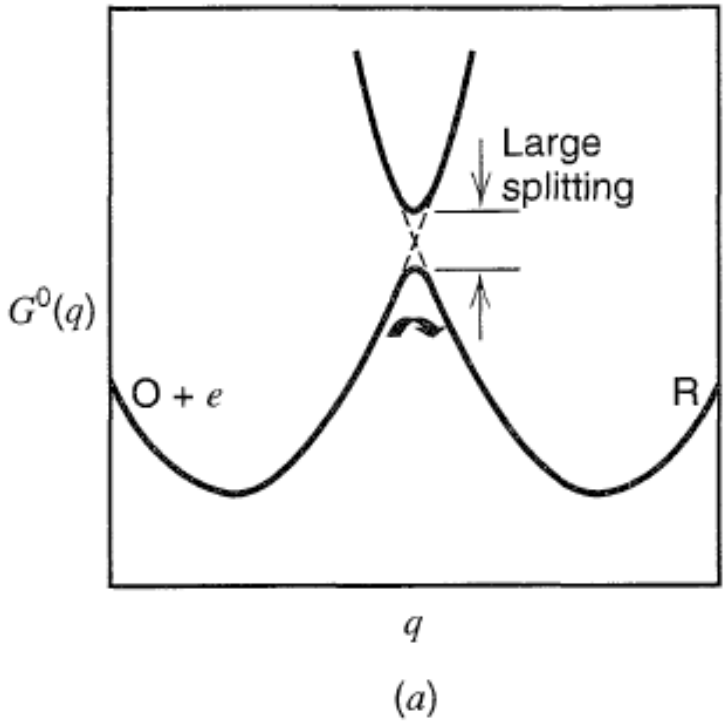
不同电子能级上的正反向电流

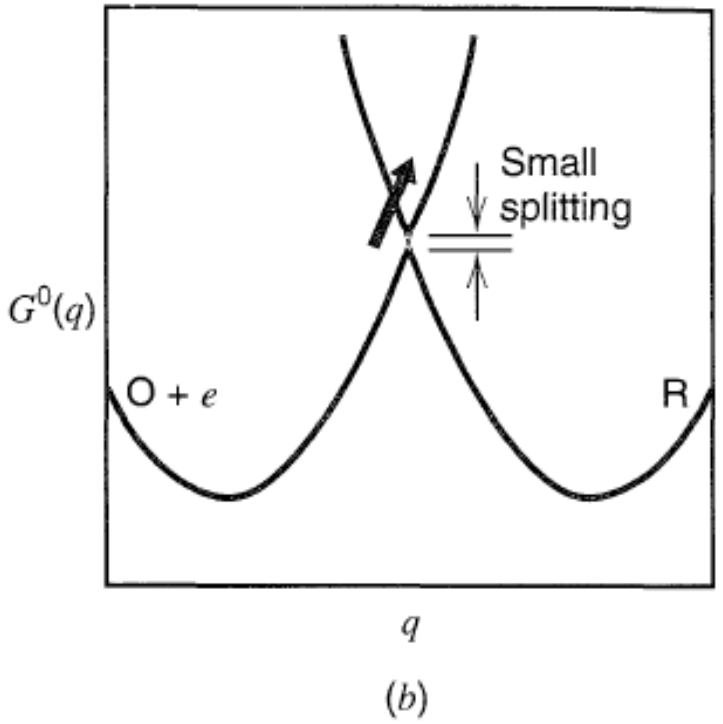
（4）电化学极化与电子能级匹配



电化学阳极极化和阴极极化以及正反电流的变化

（5）隧道效应（电子）



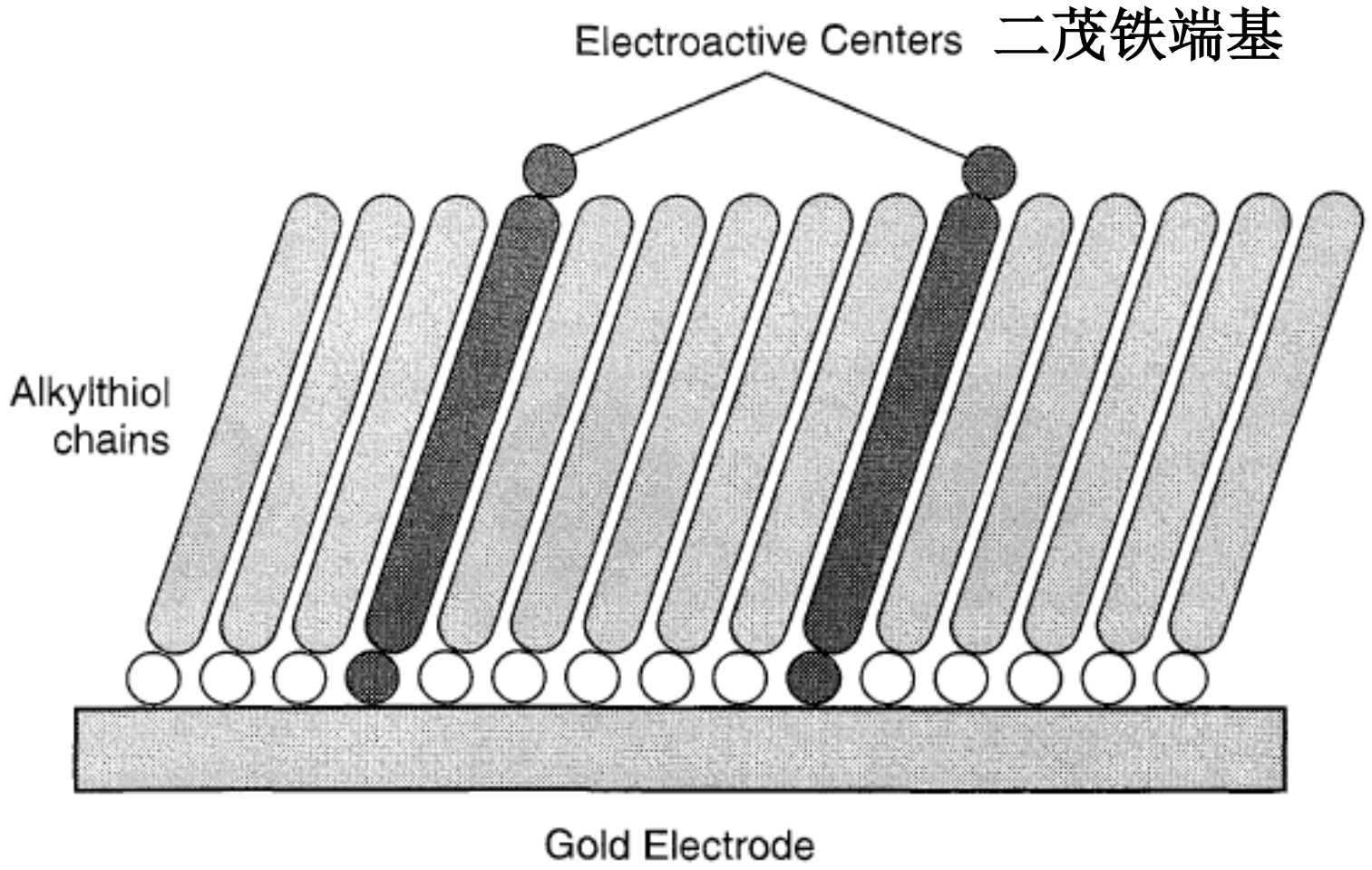


电化学电子转移行为通常看作电极的能态与反应物能态之间的电子隧穿。

$$k(x) = \kappa_{el}^{0} \exp\left(-\beta x\right)$$

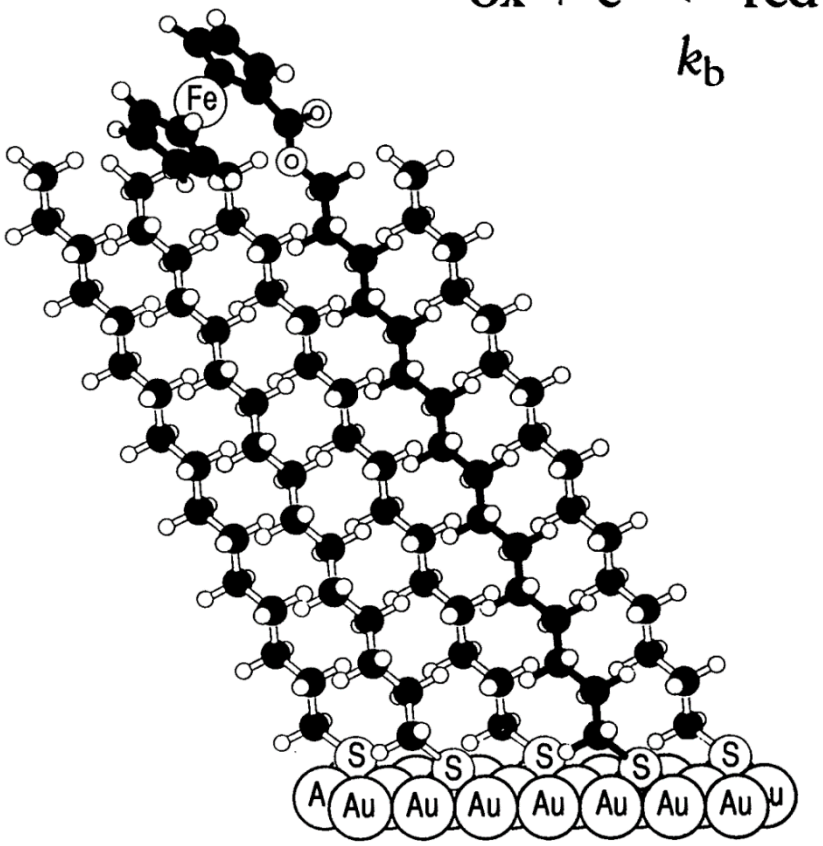
$$\beta = \frac{4\pi}{h}\left(2m\Phi\right)^{1/2}$$

（5）隧道效应（电子）


Electroactive Centers
二茂铁端基
Alkylthiol
chains
Gold Electrode

（5）隧道效应（电子）

$$\mathrm{ox} + \mathrm{e}^- \underset{k_b}{\overset{k_f}{\rightleftarrows}} \mathrm{red}$$



Christopher E. D. Chidsey，
Science, 1991, 251, 919-922.

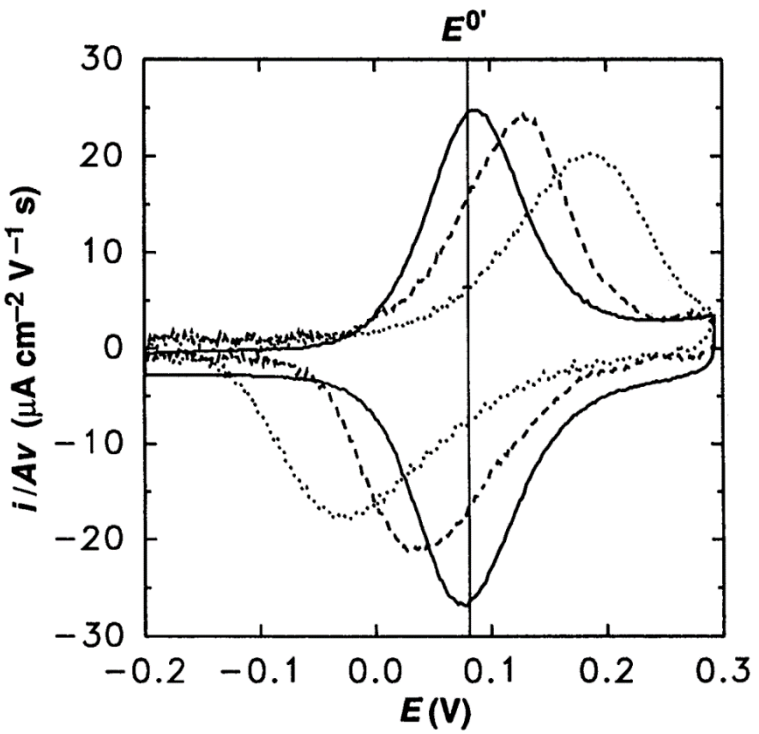


Fig. 2. Cyclic voltammograms of a mixed monolayer of ferrocene-terminated alkane thiol and unsubstituted alkane thiol on gold. The current i is normalized by the electrode area, $A = 0.70$ cm^2, and potential scan with rate v: 10 mV s^{-1} (solid), 120 mV s^{-1} (dashed), and 500 mV s^{-1} (dotted). The formal potential of the ferrocene group, $E^{0'}$, is indicated by the vertical line. Potentials are measured relative to a Ag/(1 mMAgClO$_4$, 1 M HClO$_4$) reference electrode.

（5）隧道效应（电子）

Butler-Volmer Equation

$$k_f + k_b = k^0 \exp[+\alpha e(E - E^{0'})/(k_B T)] + k^0 \exp[-(1-\alpha)e(E - E^{0'})/(k_B T)]$$

$$k_{f,b} = \sum_i f_{\pm}(\varepsilon_i)\, \upsilon_i \exp[-\Delta G^{\ddagger}_{f,b}(\varepsilon_i)/(k_B T)]$$

$$\Delta G^{\ddagger}_{f,b}(\varepsilon_i) = [\lambda \pm e(\varepsilon_i - E^{0'})]^2/(4\lambda)$$

$$k_{f,b} = \upsilon\rho(4\pi\lambda k_B T)^{1/2}\, \Omega_{f,b}(\lambda, E - E^{0'}, T)$$

$$\Omega_{f,b}\left(\lambda, E - E^{0'}, T\right) = \mathrm{C}^{1/2} \int_{-\infty}^{+\infty} g(x)\, dx$$

$$\mathrm{C} = \frac{kT}{4\pi\lambda} \qquad g(x) = \frac{\exp\left\{-\left[x - \frac{\lambda \pm e\left(E - E^0\right)}{kT}\right]^2 \frac{1}{C\pi}\right\}}{1 + \exp(x)}$$

（5）隧道效应（电子）

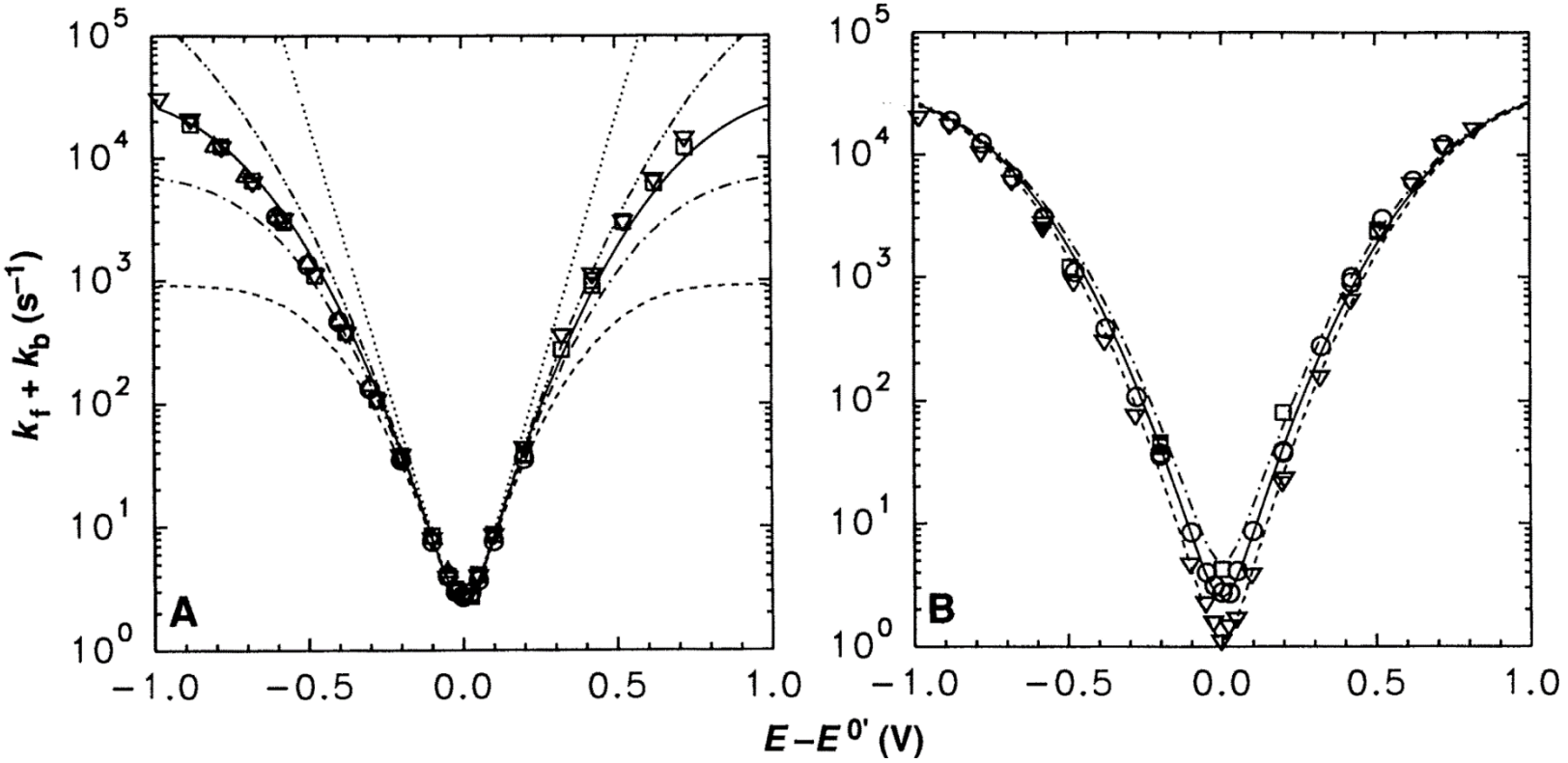


Fig. 4. (**A**) Semi-log plot of the measured decay rate constants at 25°C for four samples (symbols) and calculated decay constants from Eq. 3 with $k^0 = 1.25\ s^{-1}$, $\alpha = 0.5$ (dotted); and from Eq. 7 and 8 with: $\lambda = 1.1$ eV, $\upsilon\rho = 7.48 \times 10^5\ s^{-1}\ eV^{-1}$ (dot-dot-dashed); $\lambda = 0.85$ eV, $\upsilon\rho = 6.73 \times 10^4\ s^{-1}\ eV^{-1}$ (solid); $\lambda = 0.7$ eV, $\upsilon\rho = 1.55 \times 10^4\ s^{-1}\ eV^{-1}$ (dot-dashed); and $\lambda = 0.5$ eV, $\upsilon\rho = 2.29 \times 10^3\ s^{-1}\ eV^{-1}$ (dashed). (**B**) Semi-log plot of the measured decay rate constants at: 1°C (triangles), 25°C (circles), and 47°C (squares); and calculated decay constants from Eq. 7 and 8 with $\lambda = 0.85$ eV, $\upsilon\rho = 6.73 \times 10^4\ s^{-1}\ eV^{-1}$.

（5）隧道效应（电子）

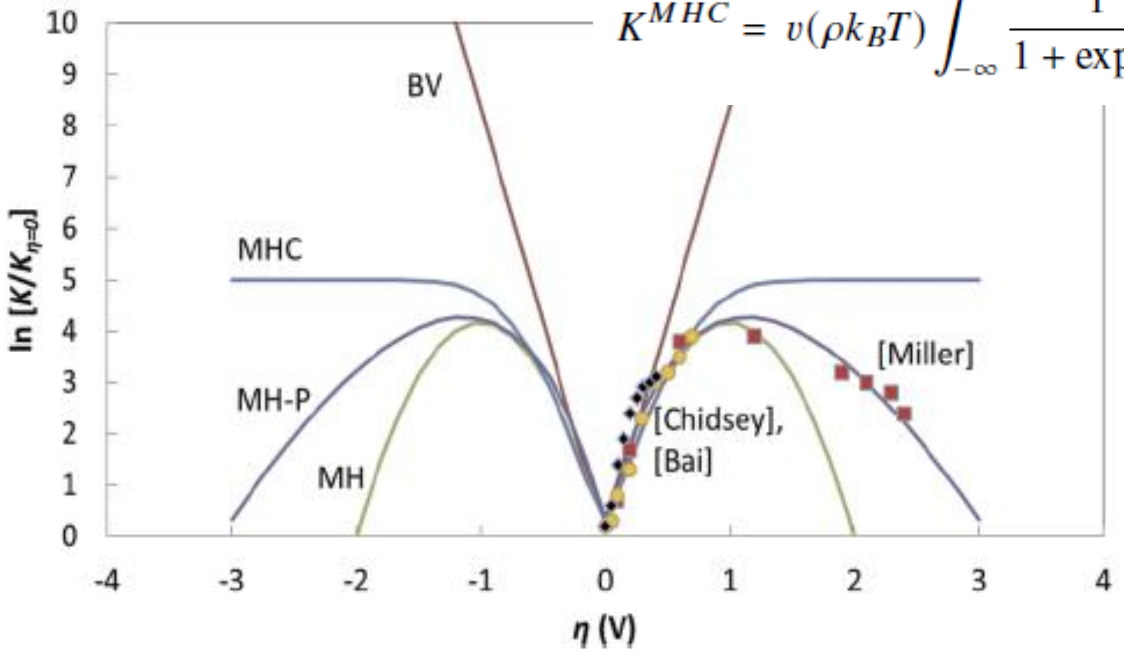
电化学反应速率常数随电极过电位的变化

Butler-Volmer公式(BV)
Marcus-Hush理论(MH)
Marcus-Hush-Chidsey理论(MHC)
Marcus-Hush-Possion拟合(MH-P)

$$K^{BV} = K_F + K_B = K^o exp\left[\frac{\alpha e\eta}{k_B T}\right] + K^o exp\left[-\frac{(1-\alpha)e\eta}{k_B T}\right]$$

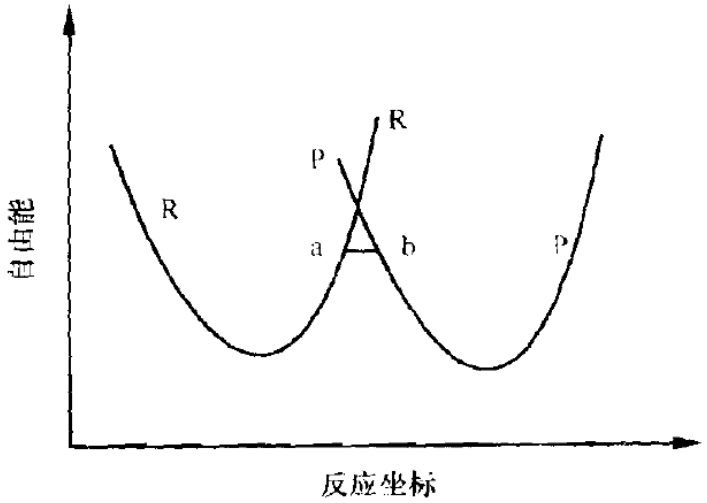
$$K^{MH} = v exp\left[-\frac{\Delta G^a}{k_B T}\right] = v exp\left[-\frac{(\lambda \mp e\eta)^2}{4\lambda k_B T}\right]$$

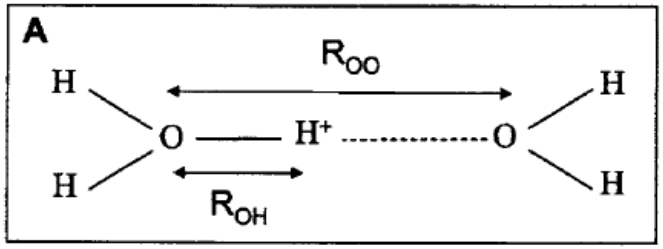
$$K^{MHC} = v(\rho k_B T)\int_{-\infty}^{\infty} \frac{1}{1+\exp(x)} exp\left(-\frac{k_B T}{4\lambda}\left[x - \frac{\lambda \mp e\eta}{k_B T}\right]^2\right) dx$$

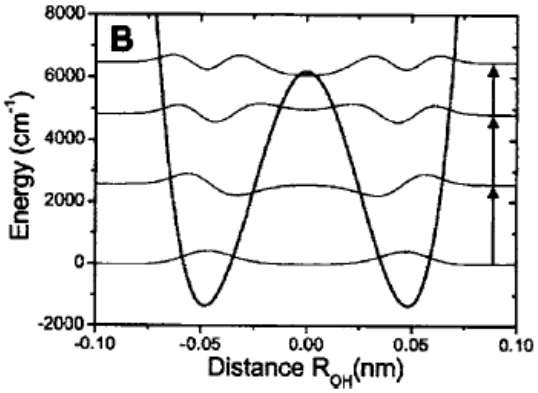


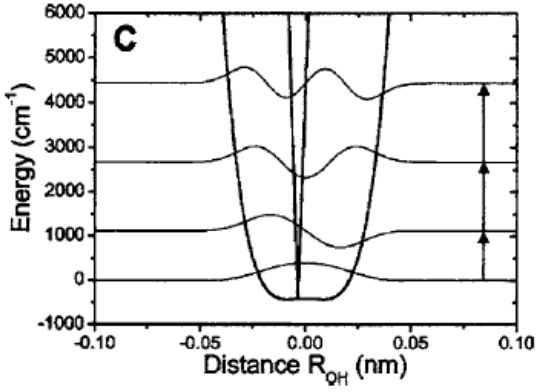
H. Yamada, P. R. Bandaru, AIP Advances, 2016, 6, 065325.

（5）隧道效应(核)









- 在电化学表面的反应有电子隧道效应，而且也可能发生原子核(H^+)甚至原子团的隧道效应，但核的隧穿距离要小。

- 温度对隧道效应影响较小。用于判断原子或原子团转移机理。

J. Chem. Phys., 2007, 126, 034511.

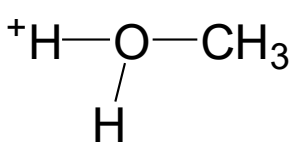
溶剂效应对电化学电荷转移的影响

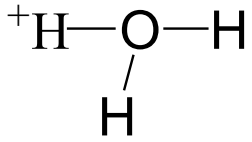
溶剂种类：水，有机溶剂，离子液体，混合溶剂…

溶剂效应：溶剂化效应、溶剂与电化学活性物种的竞争吸附、界面结构、与反应机理和电极材料性质有关有关。

1． $H_3O^+ + e \xrightarrow{rds} H_{ads} + H_2O$ Hg, Sn, Pb

CH_3OH vs H_2O 10^3 ↑





2． $H_3O^+ + H_{ads} + e \xrightarrow{rds} H_{ads} + H_2O$ Ni

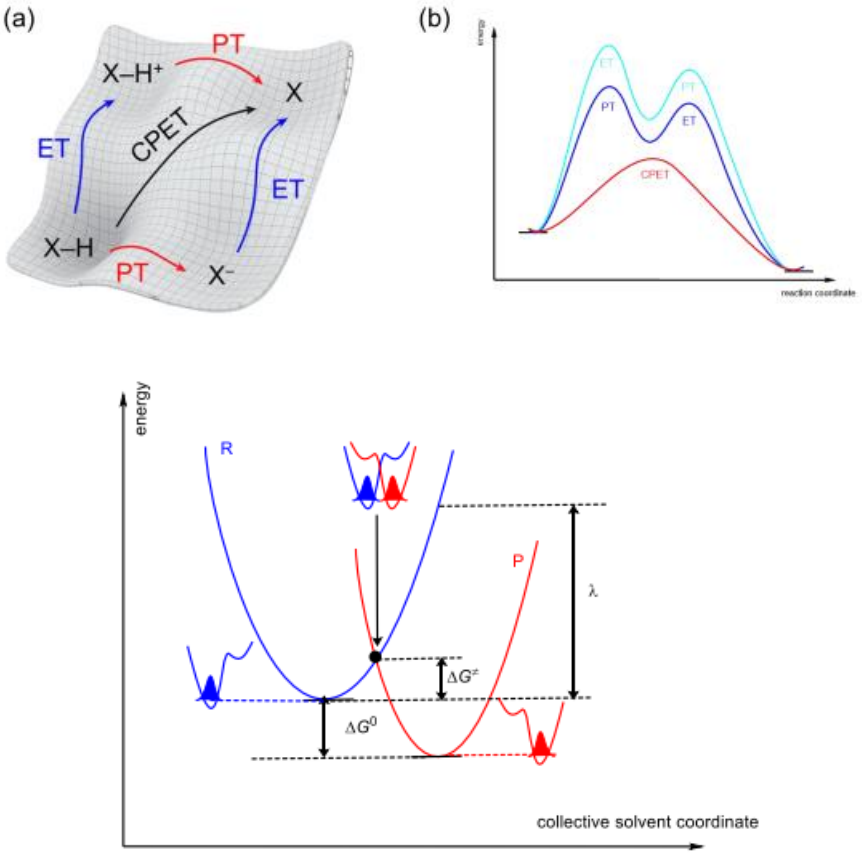
L. I. Krishtalik, Charge transfer reactions in electrochemical and chemical processes, Consulants bureau, New York, 1986.

质子耦合电子转移反应

Marcus 电子转移理论和质子转移理论

Hammes-Schiffer 等人发展了协同的质子、电子转移理论。转移的质子通过量子力学波函数表示。Marcus 理论中反应物和产物电子态的势能抛物线在CPET理论中则对应于电子态-质子振动态。

两条抛物线的交点对应 CPET 反应的非绝热过渡态，此时反应物和产物达到振动简并态，质子和电子从给体转移至受体，随着周围环境的进一步重组反应坐标沿着产物的势能抛物线直至生成最稳定的产物结构（即产物势能抛物线的最低点）。

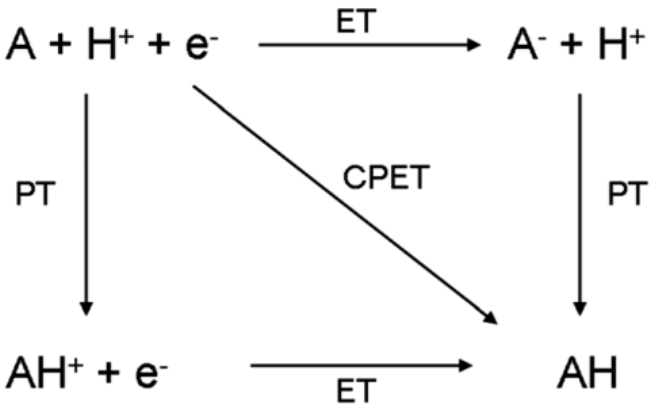


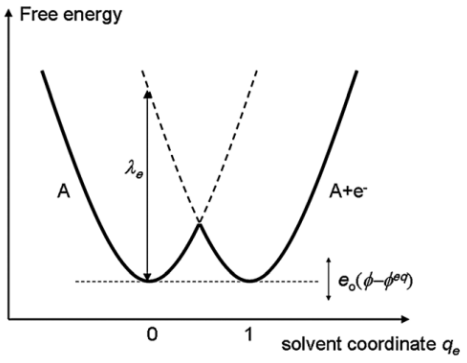
CPET 反应中反应物（R）和产物（P）非绝热振动态的势能曲线

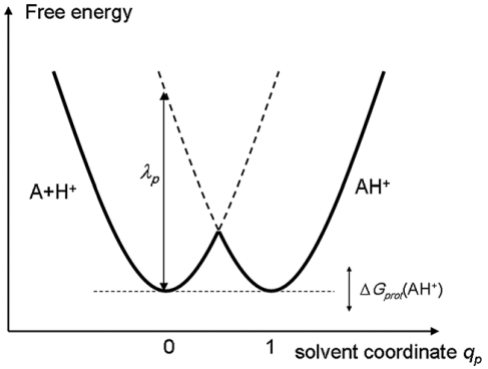
Caldin E F, Tunneling in proton-transfer reactions in solution. Chem. Rev., 1969, 69: 135-156.

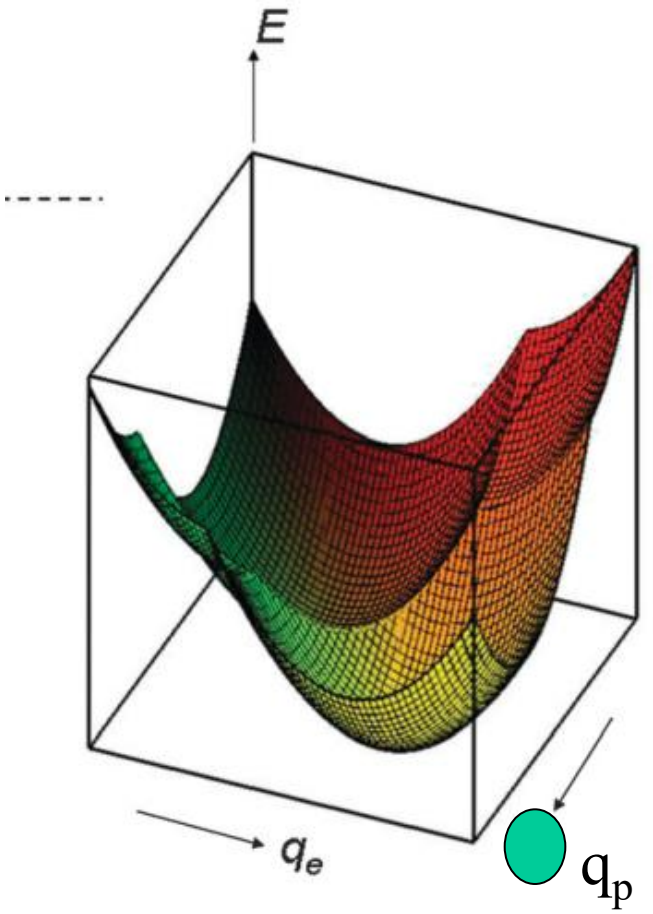
Soudackov A, Hatcher E, Hammes-Schiffer S, Quantum and dynamical effects of proton donor-acceptor vibrational motion in nonadiabatic proton-coupled electron transfer reactions. J. Chem. Phys., 2005, 122: 14505.

Theory of the transition from sequential to concerted electrochemical proton–electron transfer









Marc T. M. Koper，Phys. Chem. Chem. Phys., 2013, 15, 1399-1407
DOI: 10.1039/c2cp42369c

Quantum theory of concerted electronic and nuclear fluxes associated with adiabatic intramolecular processes，

薛定谔方程　$\mathrm{H}\Psi = E\Psi$

$$H_{\mathrm{T}} = -\frac{\hbar^2}{2m_{\mathrm{e}}}\sum_{i=1}^{N_{\mathrm{e}}}\nabla_{\mathbf{r}_i}{}^2 - \sum_{a=1}^{N_{\mathrm{n}}}\frac{\hbar^2}{2M_a}\nabla_{\mathbf{R}_a}{}^2 + V(\mathbf{r},\mathbf{R})$$

$$V = V_{\mathrm{ee}} + V_{\mathrm{en}} + V_{\mathrm{nn}}$$

$$V_{\mathrm{ee}} = \frac{e^2}{8\pi\varepsilon_0}\sum_{i=1}^{N_{\mathrm{e}}}\sum_{j\neq i}^{N_{\mathrm{e}}}\frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}$$

$$V_{\mathrm{en}} = -\frac{e^2}{4\pi\varepsilon_0}\sum_{i=1}^{N_{\mathrm{e}}}\sum_{a=1}^{N_{\mathrm{n}}}\frac{Z_a}{|\mathbf{r}_i - \mathbf{R}_a|}$$

$$V_{\mathrm{nn}} = \frac{e^2}{8\pi\varepsilon_0}\sum_{a=1}^{N_{\mathrm{n}}}\sum_{b\neq a}^{N_{\mathrm{n}}}\frac{Z_a Z_b}{|\mathbf{R}_a - \mathbf{R}_b|}$$

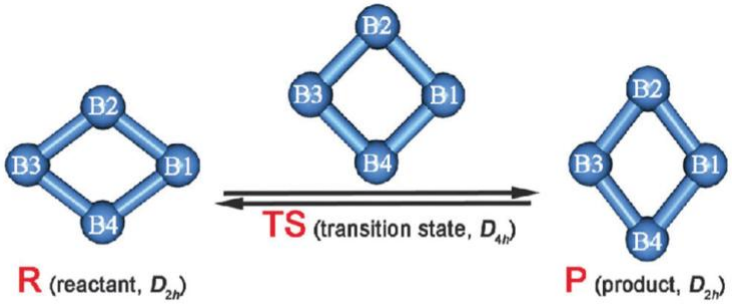
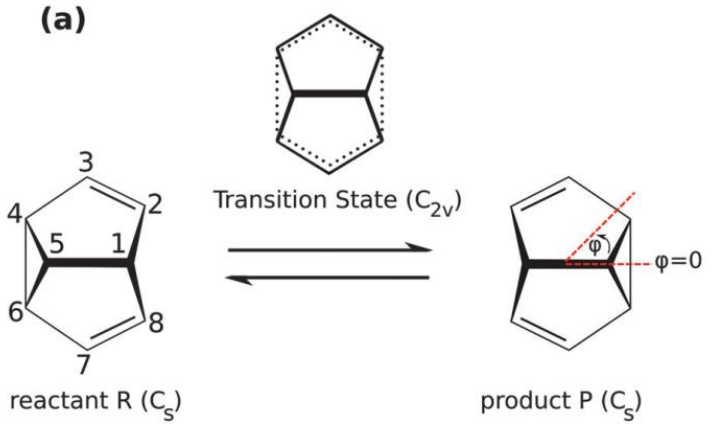
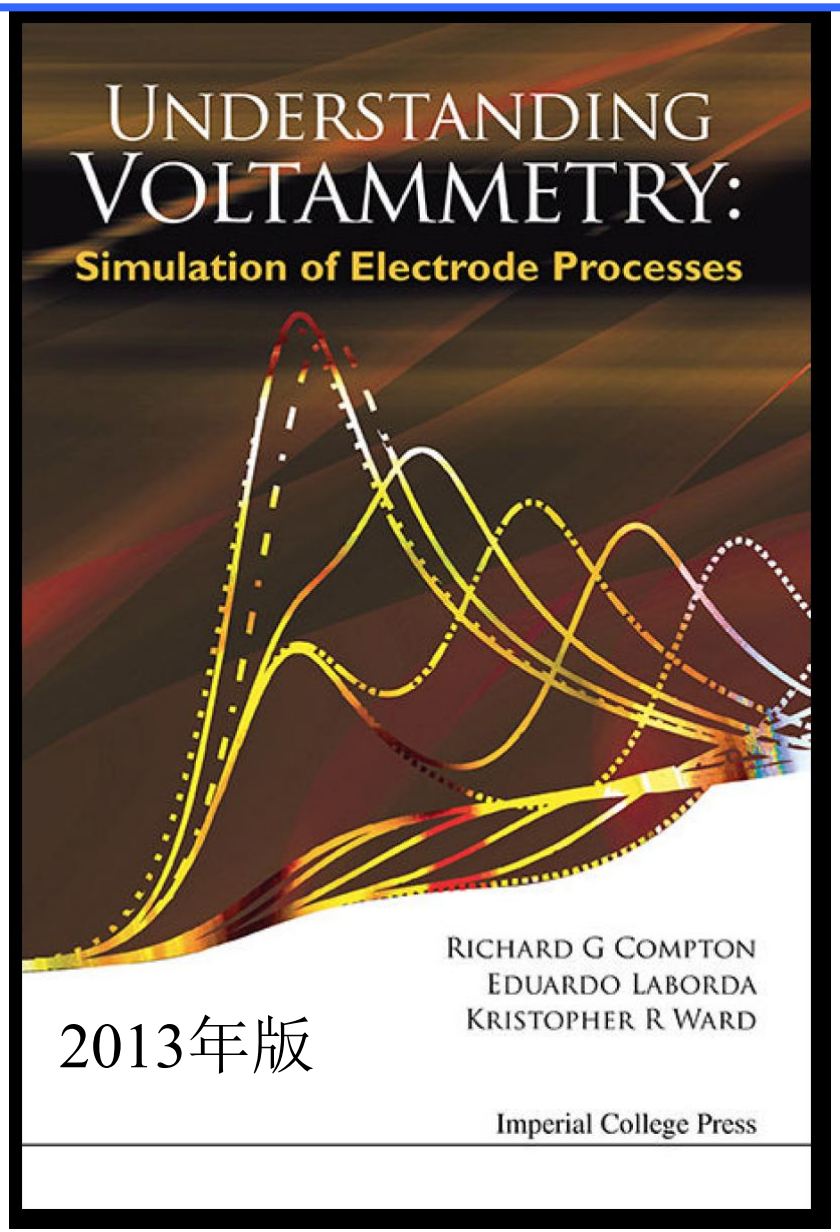


Fig. 8　Tunnelling isomerization of B_4.



T. Bredtmann, D. J. Diestler, S. D. Li, J. Manz, J. Fredy Perez-Torres, Wen-Juan Tian, Y.-B. Wu, Y. Yang and H.-J. Zhai，Phys.Chem.Chem.Phys., 2015, 17, 29421

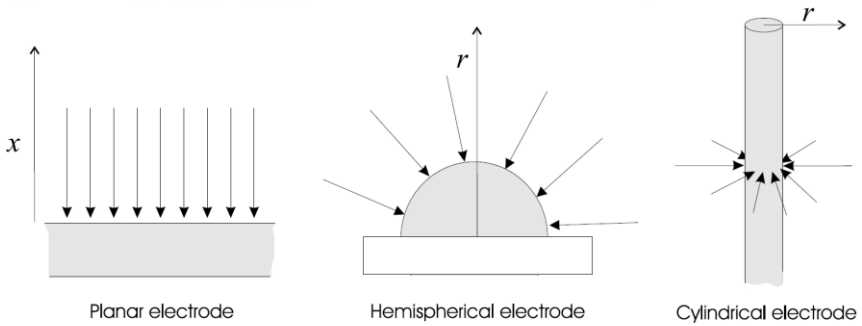
(6). 电荷转移反应的新进展



2013年版







11 Voltammetry at the Nanoscale

11.1 Transport to Particles Supported on an Electr

11.2 Nanoparticle Voltammetry: The Transport Cl
Electrode Shrinks in Size . . . . . .

11.3 Altered Chemistry at the Nanoscale . . . . .

11.4 The Electrochemical Study of Nanoparticles i
"Nano-impacts" . . . . . .

References . . . . . .

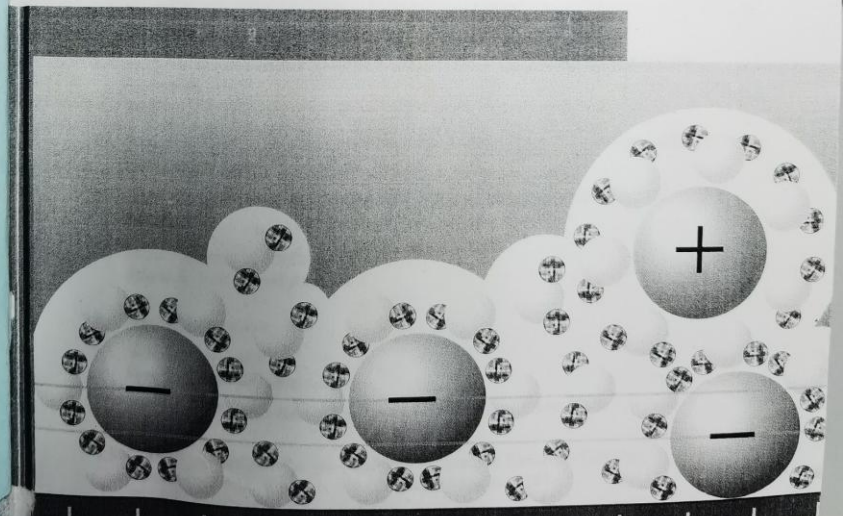
Bard–Stratmann

Encyclopedia of Electrochemistry

Volume 1

Thermodynamics and Electrified Interfaces

Volume Editors:
Eliezer Gileadi, Michael Urbakh



WILEY-VCH

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Ultra Vision LEICA Quad Camera

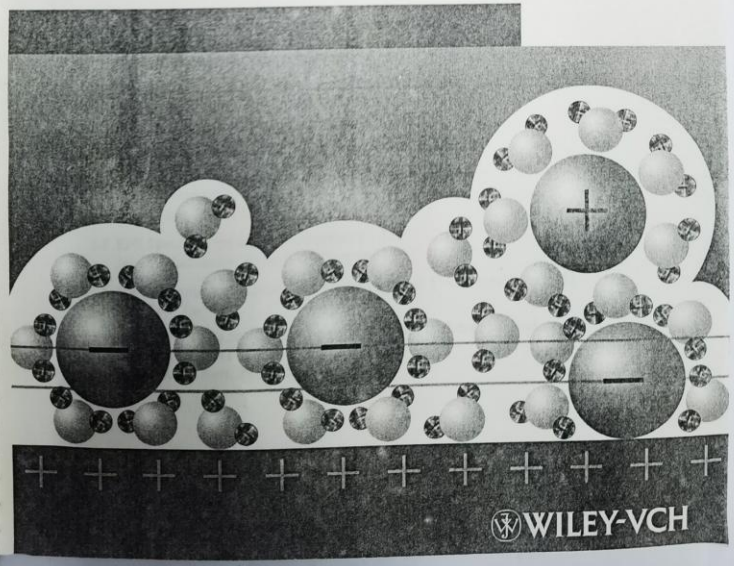
Bard–Stratmann

Encyclopedia of Electrochemistry

Volume 2

Interfacial Kinetics and Mass Transport

Volume Editor:
Ernesto Julio Calvo



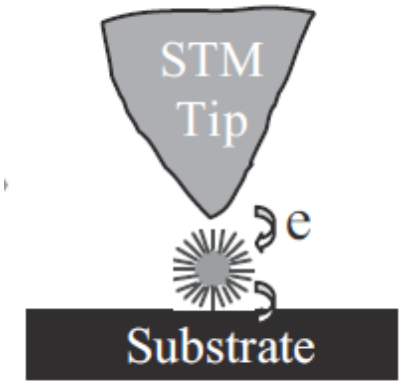
WILEY-VCH

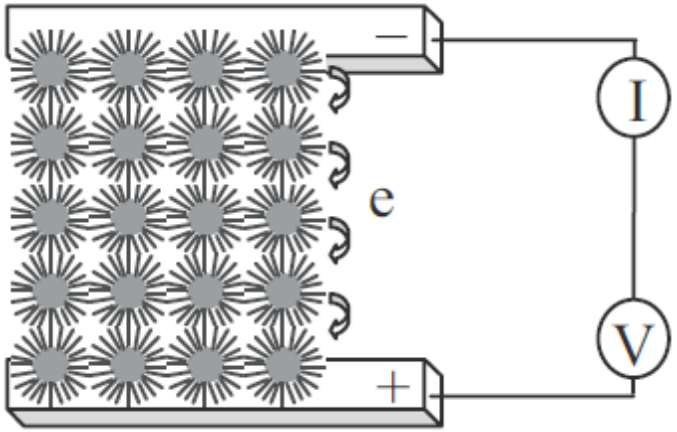
https://www.cnki.net

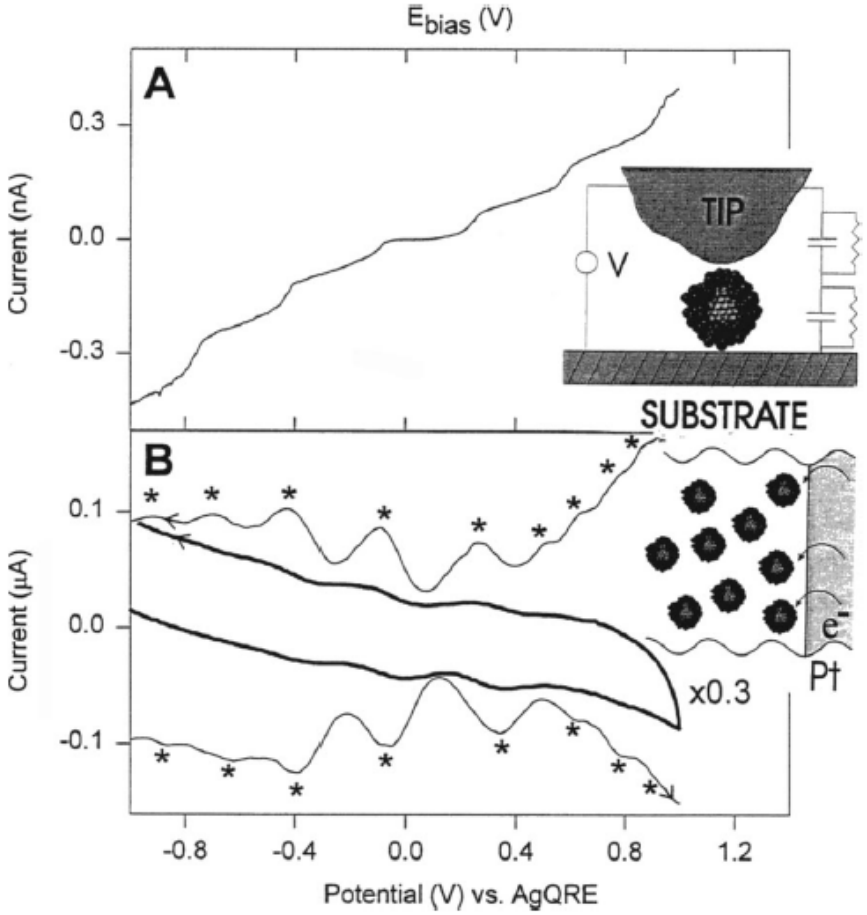
HUAWEI P40 Pro 5G
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电荷转移反应的新进展

纳米粒子固态膜中单电子的电荷转移







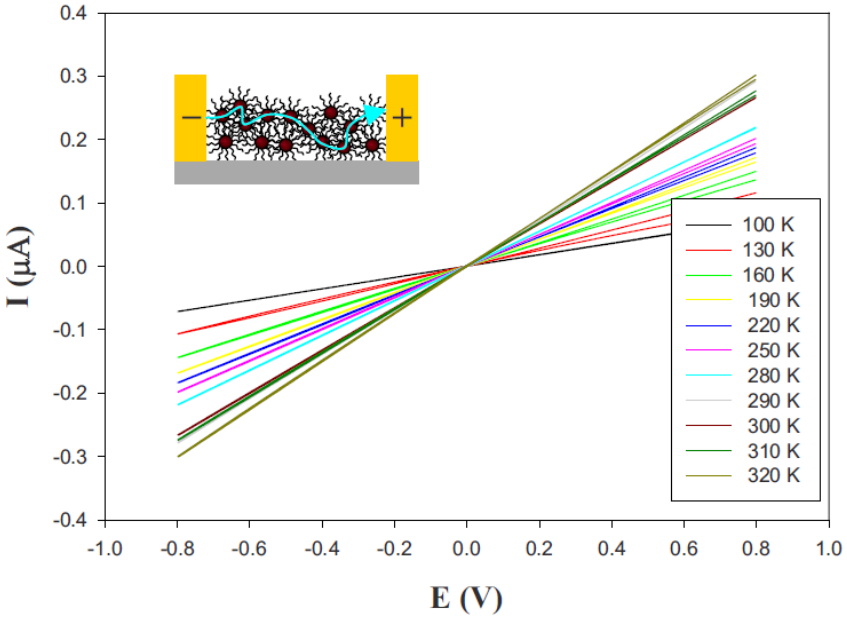
Adv. Mater. **2006**, *18*, 3279–3283

Chem. Rev. **2008**, *108*, 2688–2720

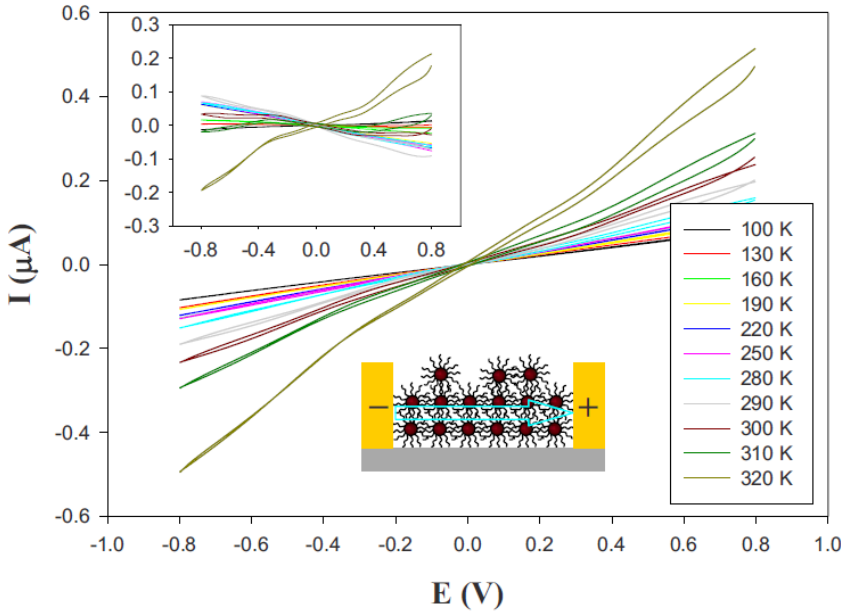
电荷转移反应的新进展

纳米粒子固态膜中单电子的电荷转移

无序纳米结构



有序纳米结构

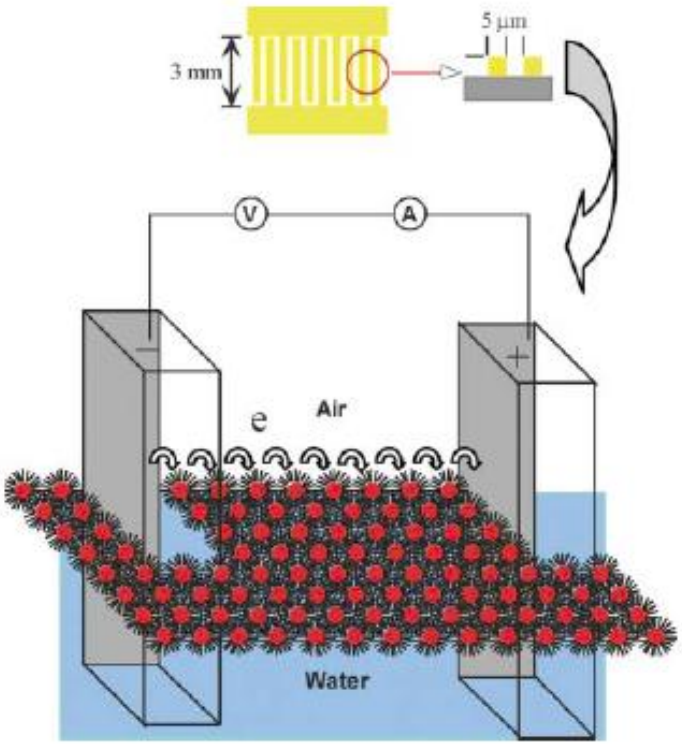


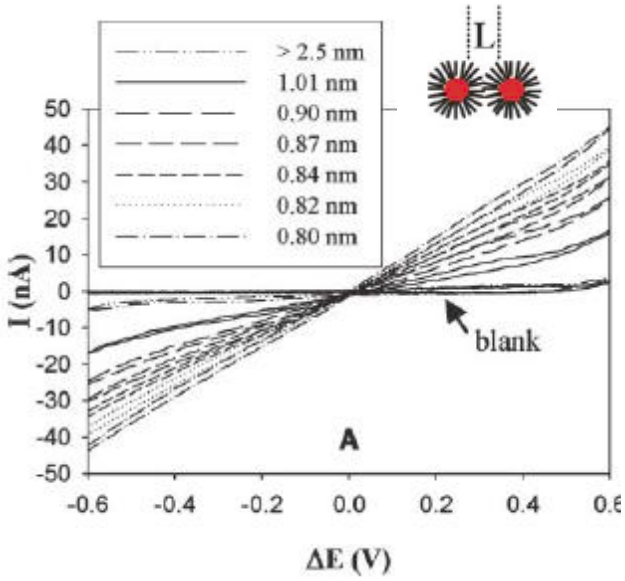
APPLIED PHYSICS LETTERS **94**, 042113 (2009)

电荷转移反应的新进展

纳米粒子固态膜中单电子的电荷转移

（single electron transfer in nanoparticle solid films）

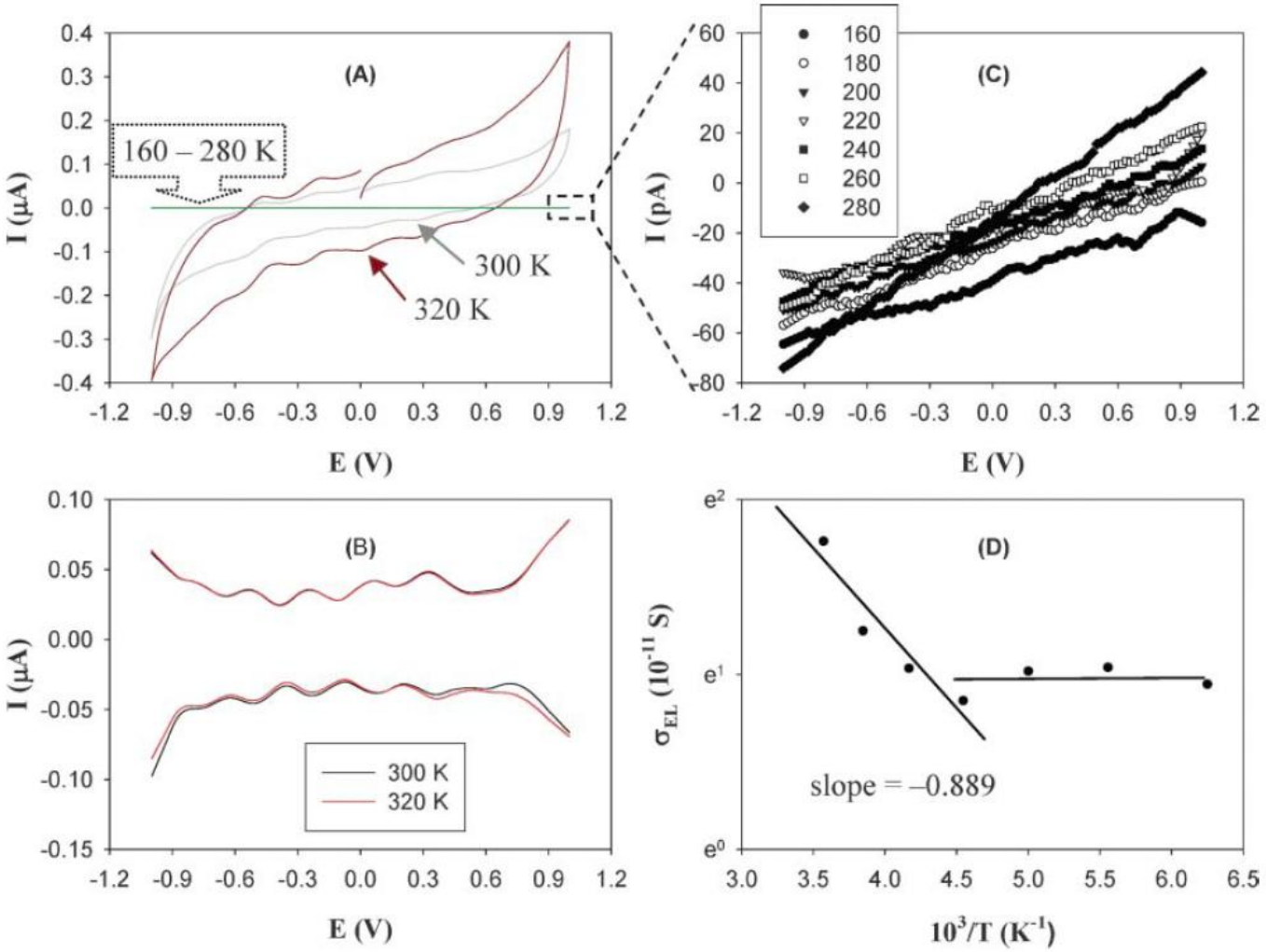




J. Mater. Chem., 2007, 17, 4115–4121

电荷转移反应的新进展

纳米粒子固态膜中单电子的电荷转移



a. 纳米粒子尺寸、形状和化学环境差异导致的无序

b. 相邻纳米粒子的偶极作用

c. 纳米粒子上的库仑排斥作用

(6). 电荷转移反应的新进展

单分子电化学(Single molecule electrochemistry)中的电荷转移

J. Am. Chem. Soc. **1996,** *118,* 9669−9675

Single Molecule Electrochemistry

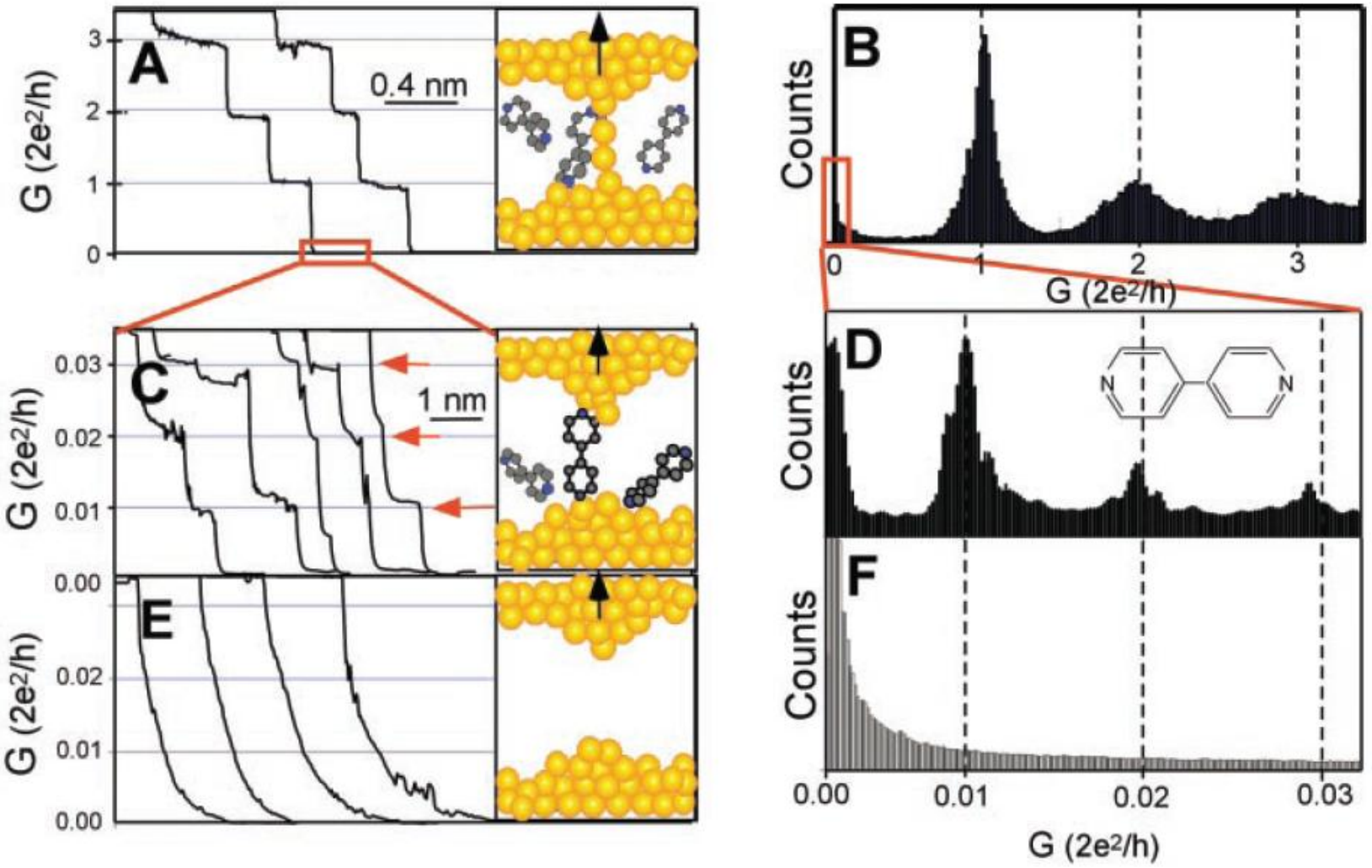
Introduction

Fu-Ren F. Fan,† Juhyoun Kwak,‡ and Allen J. Bard*,†

We report here electrochemical studies at the level of single molecules and small numbers of molecules. The detection of single molecules, their characterization, and their chemical and physical manipulation have come within current scientific capability and have become topics of significant interest. Electrochemical measurements on single molecules should allow determination of standard half-reaction potentials and free energies, diffusion coefficients, and kinetic parameters. To our knowledge the work reported here represents the first example of controlled redox chemistry at the single molecule level.

电荷转移反应的新进展

单分子电化学(Single molecule electrochemistry)中的电荷转移



BQ Xu, NJ Tao, Science, 2003, 301,1221

电荷转移反应的新进展

金属纳米粒子的电化学电荷转移

$$2(CH_3)_2C^{\bullet}OH + Ag^{x-} \rightarrow (CH_3)_2CO + H^+ + Ag^{(x+2)-}$$

$$Ag^{(x+2)-} + 2H^+ \rightarrow Ag^{x-} + H_2 \qquad k = 5\text{x}10^{-17}\ \text{cm}^{-2}\ \text{s}^{-1}$$

over millisecond and minute time scales, respectively. From the charge storage properties of the Ag nanoparticles, it was estimated that, under the irradiation conditions used, an approximately 4 nm diameter Ag nanoparticle could carry as many as 450 stored electrons, with a double layer capacity of 43 μF/cm^2 of the nanoparticle surface. Calculations indicated that as many as 300 electrons could persist on a nanoparticle for about 1 min after irradiation had ceased.

Chem. Rev. **2008**, *108*, 2688–2720

(6). 电荷转移反应的新进展

Abraham Nitzan

1.3 Electronic Tunnel Factors in Molecular Electron Transfer and Molecular Conduction

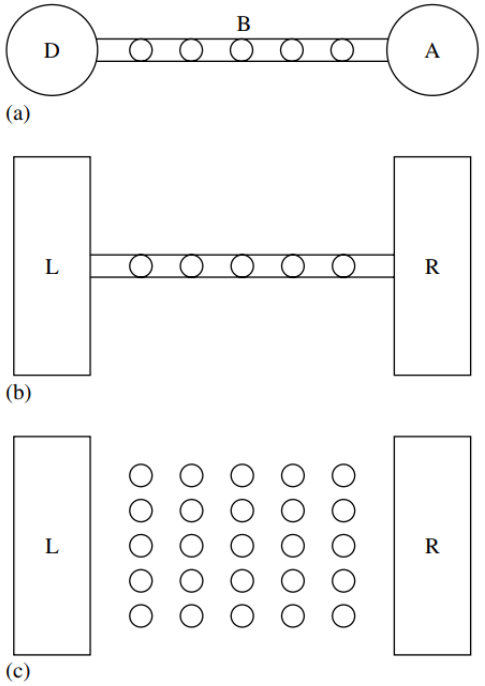


Fig. 1 Schematic views of typical electron transmission systems. (a) A "standard" electron transfer system containing a donor, an acceptor, and a molecular bridge connecting them (not shown are nuclear motion baths that must be coupled to the donor and acceptor species). (b) A molecular bridge connecting two electronic continua, Land R, representing, e.g., two metal electrodes (c) same as (b) with the bridge replaced by a molecular layer.

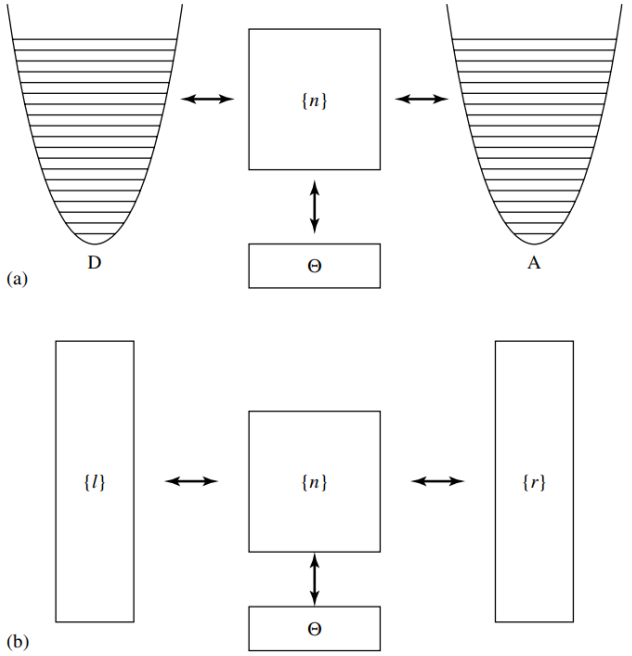


Fig. 2 A schematic view of the electronic and nuclear states involved in typical electron transmission systems.

(6). 电荷转移反应的新进展

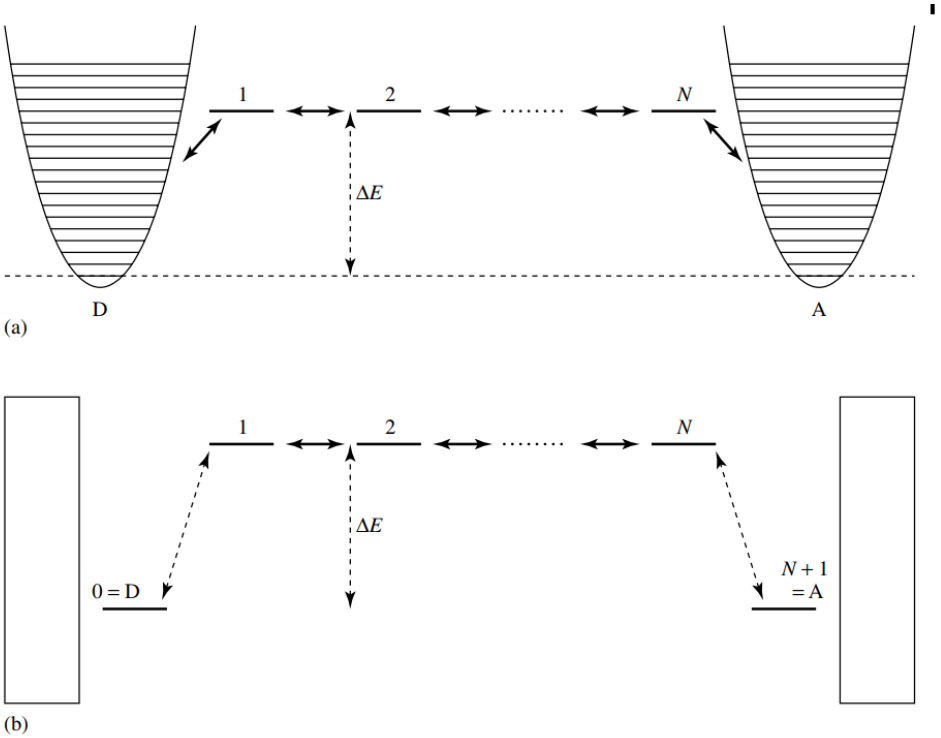
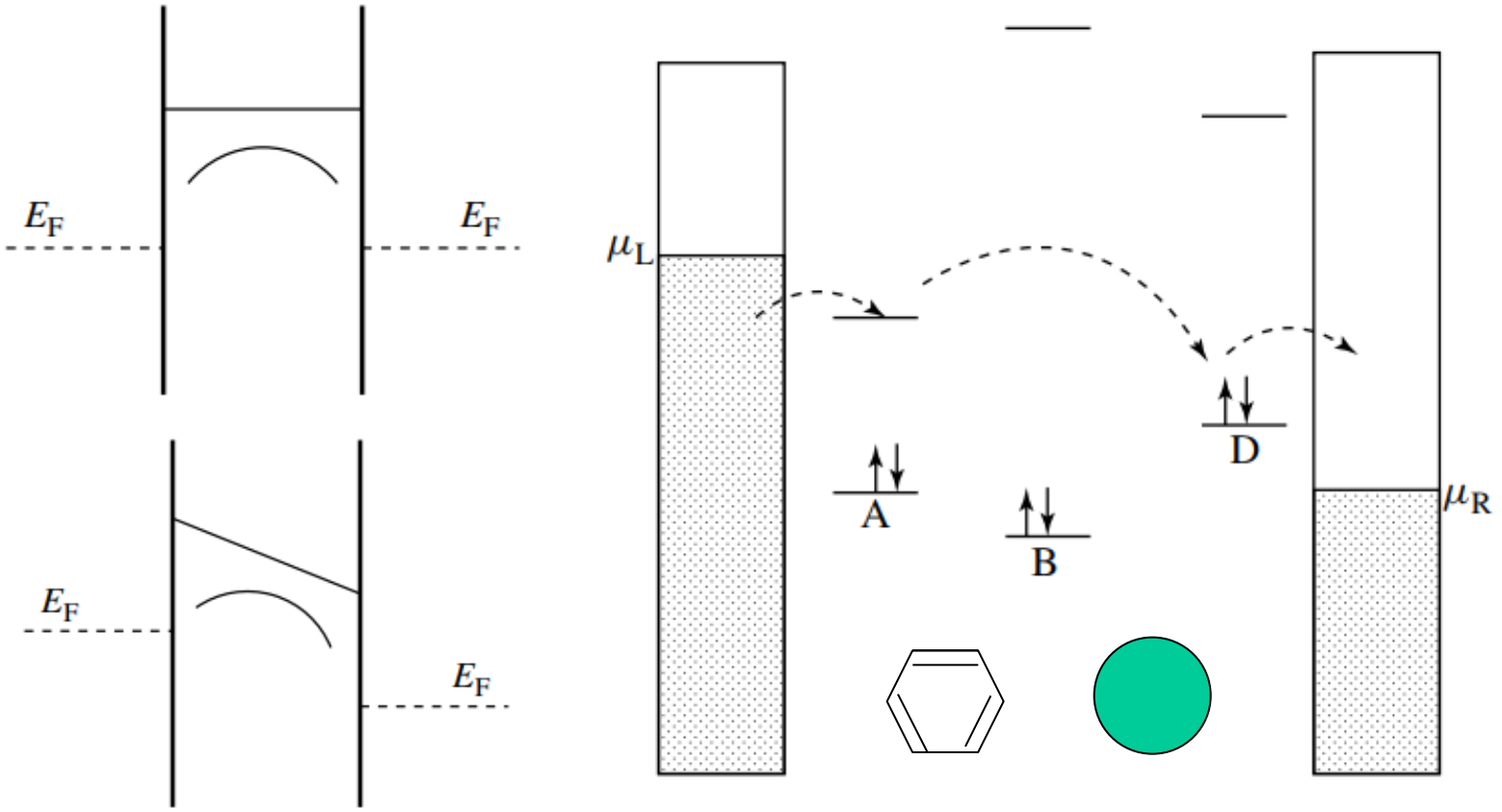


Fig. 3 Simple level structure models (a) for molecular electron transfer and (b) for electron transmission. The molecular bridge is represented by a simple set of levels that represent local orbitals of appropriately chosen bridge sites. This set of levels is coupled to the donor and acceptor species (with their corresponding nuclear environments) in (a), and to electronic continua representing metal leads (say) in (b). In the latter case, the physical meaning of states 0 and N + 1 depends on the particular physical problem: They can denote donor and acceptor states coupled to the continua of environmental states (hence the notation 0 = D, N + 1 = A), surface localized states in an metal-molecule-metal junction, or they can belong to the right and left scattering continua.

(6). 电荷转移反应的新进展



A model for current rectification in a molecular junction: Shown are the chemical potentials μL and μR in the two electrodes, and the HOMO and LUMO levels of the donor, acceptor, and bridge. When the right electrode is positively biased, (as shown) electrons can hop from left to right as indicated by the dotted arrows.

电荷转移反应的新进展

热电子电化学中的电荷转移

4.2.1. Hot electron electrochemistry

'Hot electrons' are electrons excited inside a metallic or semiconductor phase [123]. They are thermally far from equilibrium but accord with a modified Fermi distribution function with a higher 'effective' temperature $T^* > T$. Hot electrons are broadly important in solid state photo-excitation, photo-excited interfacial charge transfer, energy photoconversion, and solid state tunnel junctions. The latter has been extended to the metal–electrolyte interface [20,21].

AM Kuznetsov, J. Ulstrup, Electrochim. Acta, 2000, 45, 2339-2361

思考题 6（2026.10.22）

1. 在**Marcus**电子转移理论中电子转移重组能包括内重组能和溶剂重组能，若电解质溶液为水，分子半径为**4.0Å**，距电极表面**7Å**，考虑单电子转移时，试以**eV**为单位计算溶剂重组能和活化能值。

2. 下图是金属电解质溶液界面的能级相对位置，电解质溶液中含有氧化还原对的高斯态密度分布。试写出图中相关物理量E_F, f(E), E^0, λ, $D_O(\lambda,E)$, $D_R(\lambda,E)$ 的表达式，并简述其物理意义。

